Honeywell

Honeywell TC Thermostat Configuration Wizard

USER GUIDE

31-00490-03

ABOUT THIS GUIDE

This document provides instructions to configure the TC Thermostat Configuration Wizard connected to the Niagara workbench.

To take full advantage of the information in this guide, users must have training or experience working with the Niagara N4 software.

PREREQUISITE

Table 1 Prerequisite

Parameter	Description
Software requirement	Niagara workbench. For compatibility information, refer to the latest software releaser notes.
	Google Chrome web browser
Modules	• honeywellTCThermostatWizard-rt.jar • honeywellTCThermostatWizard-ux.jar • honeywellTCThermostatWizard-wb.jar • honeywellBacnetDeviceManager-rt.jar • honeywellBacnetDeviceManager-wb.jar • honeywellDeviceManager-rt.jar • honeywellDeviceManager-wb.jar
Unit Settings	The wizard supports Imperial and Metric measurement units. To change the units in the wizard, follow the below steps. 1. Navigate to the user service of the station. 2. Double-click on the user account and configure the **Unit Conversion** property. If the **Unit Conversion** is set to **None**, the Imperial units are applied.

INTRODUCTION

The Honeywell TC Thermostat Configuration Wizard is a utility that provides a systematic approach for configuring the various operational sequences and settings for the TC500A and TC300B thermostats. The wizard configures a TC500A thermostat for conventional, Air source, Water Source heat pump and Fan coil unit and a TC300B thermostat for Fan coil unit.

It can be accessed using a workbench or web browser. For more details, refer to the TC Thermostat Configuration Wizard User Guide - 31-00490.

NOTE:

Before proceeding with the configuration upload or download, update the thermostat firmware to version TC500A = 01.01.05.00 and TC300 = 01.02.09.08 or higher to ensure compatibility between the wizard and the thermostat.

Launching the Wizard

Follow the instructions below to launch the TC500A thermostat configuration wizard:

1. Navigate to the TC500A device, click on **Station > Config > Drivers > BacnetNetwork > TC500A**.
2. Double-click on TC500A to launch the TC500A Thermostat configuration wizard application.

Schedules

TC500A supports the schedule for occupancy and purge modes.

Table 2 Schedules

Schedule Name	Description	Type	BACnet Object ID
OccSchedule	Monday - Friday (6:00 AM TO 6:00 PM) **0**: Occupied **1**: Unoccupied **3**: StandBy **255**: Null	Enum Schedule	2
Purge Schedule	**0**: PurgeOff **1**: PurgeOn The default is PurgeOff for all 5 days.	Enum Schedule	2

The schedules are added to the TC500A device by default. Configure the Occupancy and **PurgeControlSchedule** to change the modes and schedule the time. If a global schedule is present, link it to the **OccSchedule** and the **PurgeSchedule** using the **BacnetScheduleExport** component.

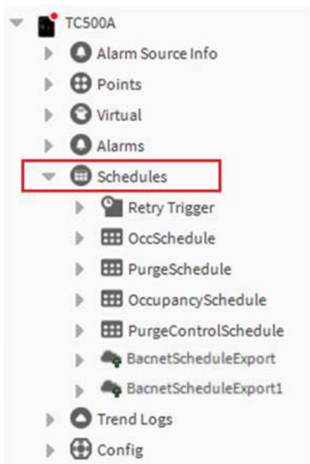

Fig. 1 Schedules

Follow the below steps to link the global schedule instead of the given occupancy schedule:

1. Navigate to the **BacnetScheduleExport** component and update the supervisor ord with the requiredschedule.

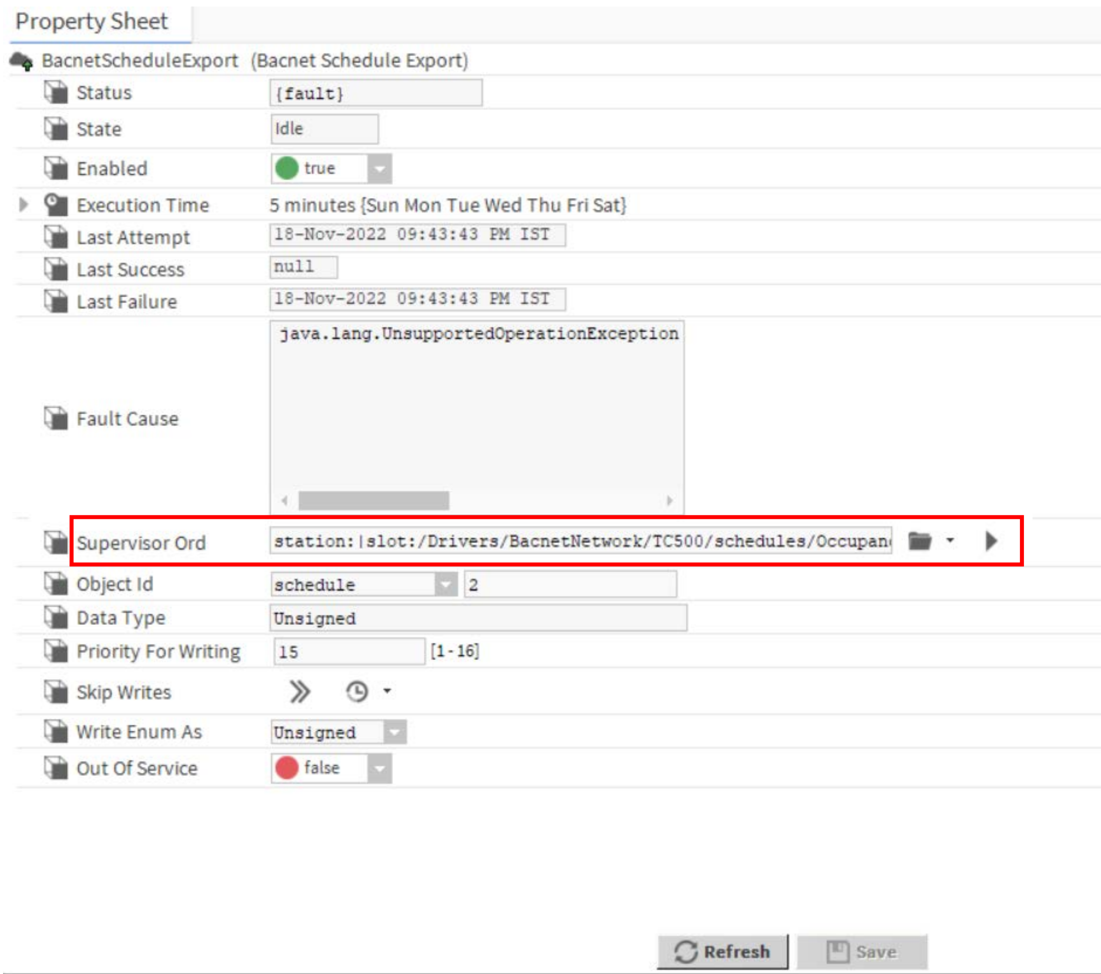

Fig. 2 BacnetScheduleExport Property Sheet

2. Click Save

Field Description

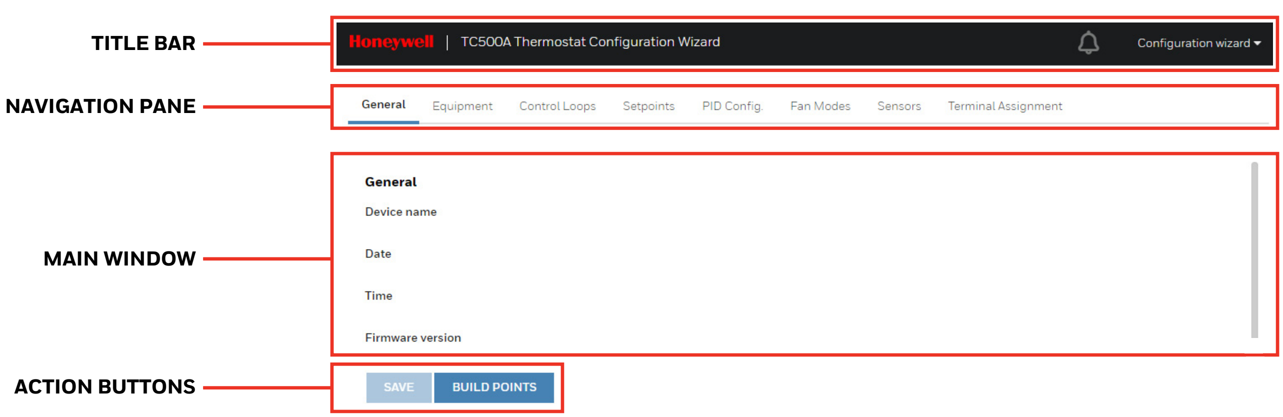

Fig. 3 TC500A Thermostat Configuration Wizard

TC Thermostat Configuration Wizard Application View

Title Bar

Displays the application name as **TC Thermostat Configuration Wizard**.

Navigation Pane

Enables the user to navigate through the various configurations of the equipment, sensors, PID control, etc.
See *Navigation Menu on page 8*

Main Window

It provides configuration settings based on the chosen parameters.
See *General - TC500A on page 9*

Action Buttons

- **SAVE**: Enables the user to save the changes made to a configuration.
- **BUILD POINTS**: Enables the user to build the points in the points folder of the TC500 device. See Table 2 below for the list of points built after clicking on the build points button.

NOTE:
Build Points are not applicable for TC300 Thermostat.

Table 3 Build Points

Point Name	Range	Type	BACnet Object ID
Common Points for Conventional and Heat Pump Applications			
no_EffSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Output	5
no_ActiveHeatStages	0 to 3	BACnet Numeric Output	7
no_ActiveCoolStages	0 to 3	BACnet Numeric Output	11
no_DaTemp	35 to 165 °F (1 to 73 °C)	BACnet Numeric Output	13
no_OaTemp	-40 to 150 °F (-39 to 65 °C)	BACnet Numeric Output	16
no_SpaceTemp	-40 to 140 °F (-39 to 60 °C)	BACnet Numeric Output	18
no_SpaceHumidity	0 to 100 %	BACnet Numeric Output	19
Cfg_Setpoints_OccCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	4
Cfg_Setpoints_UnOccCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	6
Cfg_Setpoints_OccHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	7
Cfg_Setpoints_UnOccHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	9
Cfg_Thermostat_ClAdjStPt	-30 to 30 °F (-34 to -1 °C)	BACnet Numeric Value	256
Cfg_Thermostat_HtAdjStPt	-30 to 30 °F (-34 to -1 °C)	BACnet Numeric Value	257
no_OccupancyState	**0:** Unoccupied **1:** Occupied	BACnet Boolean Output	3
no_FanStart	**0:** Off **1:** On	BACnet Boolean Output	19

Table 3 Build Points (Continued)

Point Name	Range	Type	BACnet Object ID
no_EffTempMode	**1:** Cool Mode **2:** Reheat Mode **3:** Heat Mode **4:** Emergency Heat **5:** Off	BACnet Enum Output	6
no_EffOccState	**1:** Occupied **2:** Unoccupied **3:** Bypass **4:** Standby **5:** Null	BACnet Enum Output	20
ni_OccManCom	**1:** Occupied **2:** Unoccupied **3:** Bypass **4:** Standby **5:** No Override	BACnet Enum Value	4
Cfg_Thermostat_SysSwitch	**1:** Auto **2:** Cool **3:** Heat **4:** EmergHeat **5:** Off	BACnet Enum Value	8
Cfg_Setpoints_StbyCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	5
Cfg_Setpoints_StbyHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	8
ni_SpaceCO2	0 to 2000 ppm	BACnet Numeric Value	81
ni_OutsideHum	0 to 100 %	BACnet Numeric Value	194
Additional Points for Heat Pump Applications			
no_ReversingVlv	**0:** Close **1:** Open	BACnet Boolean Output	16
no_ActiveCompHeatStages	0 to 3	BACnet Numeric Output	10
no_ActiveAuxHeatStages	0 to 2	BACnet Numeric Output	8

CONFIGURATION

Navigation Menu

General

The General Menu allows the user to configure the miscellaneous settings for the TC500A thermostat.
See *General - TC500A on page 9* and See *General - TC300 on page 10*

Equipment

This Menu allows the user to select the equipment type and subtypes. Any configuration change will generate a unique template number that determines the points to be built when the Build Points button is clicked.
See *Equipment - TC500A on page 11* and See *Equipment - TC300 on page 12*

Control Loops

This Menu allows the user to select the Economizer status, Damper positions, CO2 control, Humidification and Dehumidification. Any change in configuration results in heating and cooling. Click the build points button to generate the points associated with the template number. See *Control Loops - TC500A on page 13*.

NOTE:
Applicable only for TC500A Device.

Setpoints

This Menu allows the user to configure the setpoints for different modes like occupied, standby, and unoccupied.
See *Setpoints - TC500A on page 14* and See *Setpoints - TC300 on page 15*

PID Configuration

This Menu allows the user to configure the PID parameters for heating and cooling modes.
See *PID Configurations - TC500A on page 16* and See *PID Configurations - TC300 on page 17*

Fan Modes

This Menu allows the user to configure the different fan modes, speed types, and other fan settings.
See *Fan Modes - TC500A on page 18* and See *Fan Type - TC300 on page 19*

Sensors

This Menu allows the user to configure the different types of sensors present in a system as well as the main control sensor. See *Sensors - TC500A on page 20* and See *Sensors - TC300 on page 21*

Terminal Assignment

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings. See *Terminal Assignment - TC500A on page 22* and See *on page 23*

GENERAL - TC500A

The general Menu displays the thermostat properties such as the Device name, Location, Date, Time, Firmware version, Wi-Fi signal strength, MSTP header CRC error count, and MSTP data CRC error count. It also has miscellaneous properties such as the Power up delay time, Override time, and Demand response setpoint adjustment.

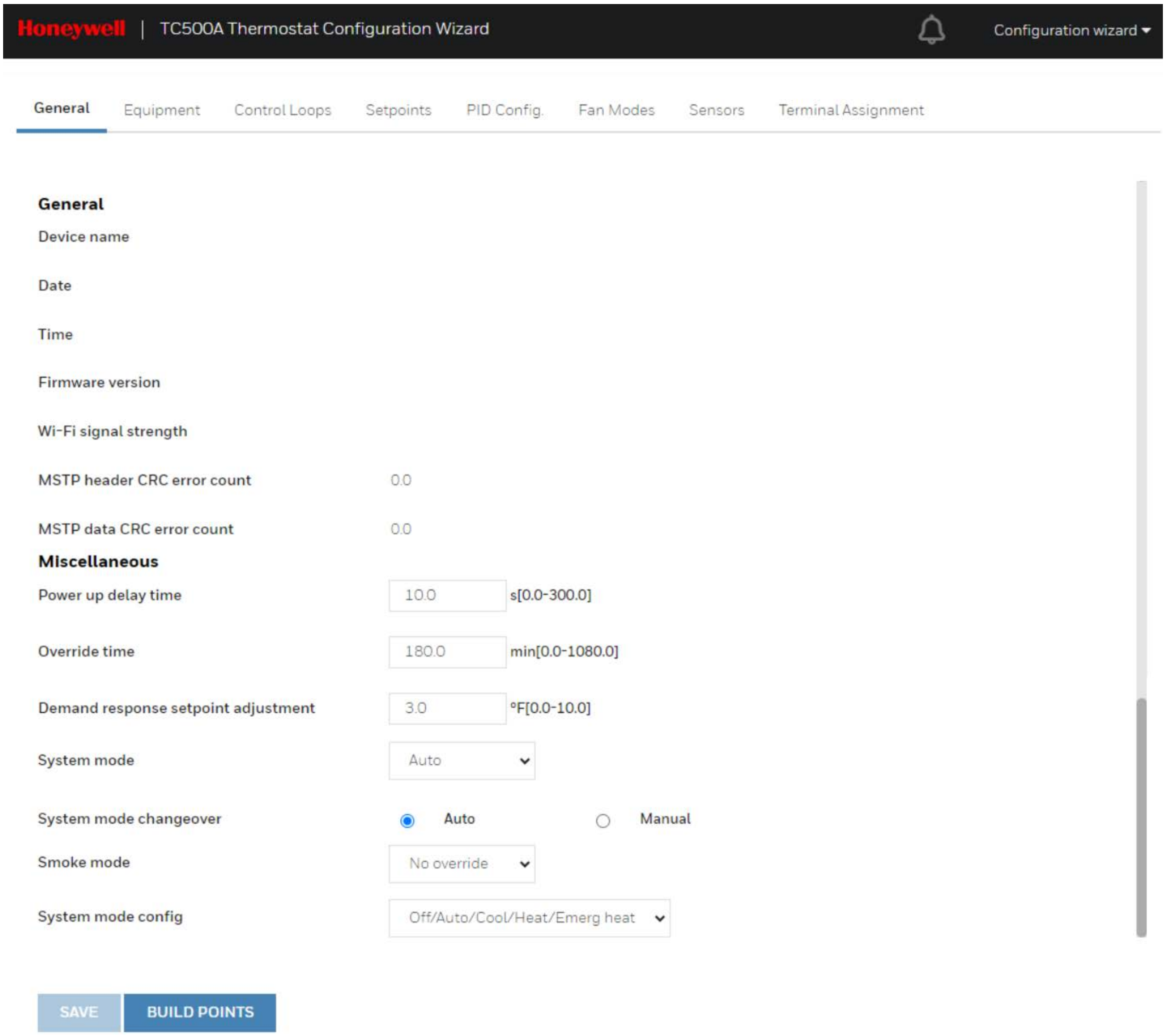

Fig. 4 General Menu View

NOTE:
Click **SAVE** for the configuration changes to reflect on the wizard. If the device is offline, not all of the parameters are displayed.

GENERAL - TC300

The general Menu displays the thermostat properties such as the Firmware Revision, MSTP header CRC error count, and MSTP data CRC error count, MSTP CRC Error Reset. It also has miscellaneous properties such as the Power up delay time, Override time, and Unoccupied Override Time, System Mode.

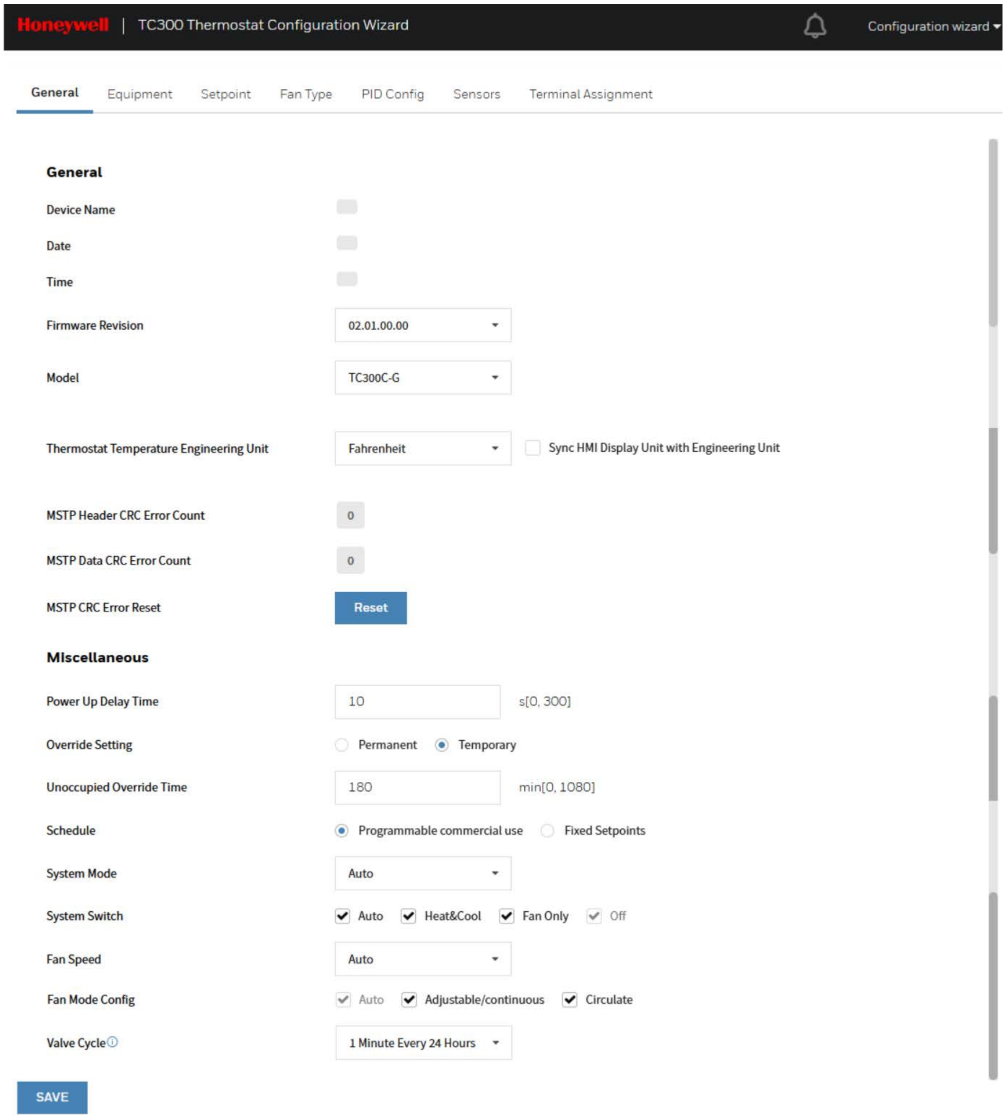

Fig. 5 General Menu View

NOTE:

Click **SAVE** for the configuration changes to reflect on the wizard. If the device is offline, not all of the parameters are displayed.

EQUIPMENT - TC500A

This Menu allows the user to select the equipment type and subtype. Any change in configuration results in generating an unique application selection number. Click the build points button to generate the points associated with the template number.

Equipment Type

Select the required equipment type through this option. It is a primary setting in the configuration as selecting the various other parameters from the different locations depends on the equipment type. The additional settings for the equipment will be enabled or disabled according to the equipment type selected.

The following equipment types are available:

- Conventional
- Air source heat pump
- Water source heat pump.

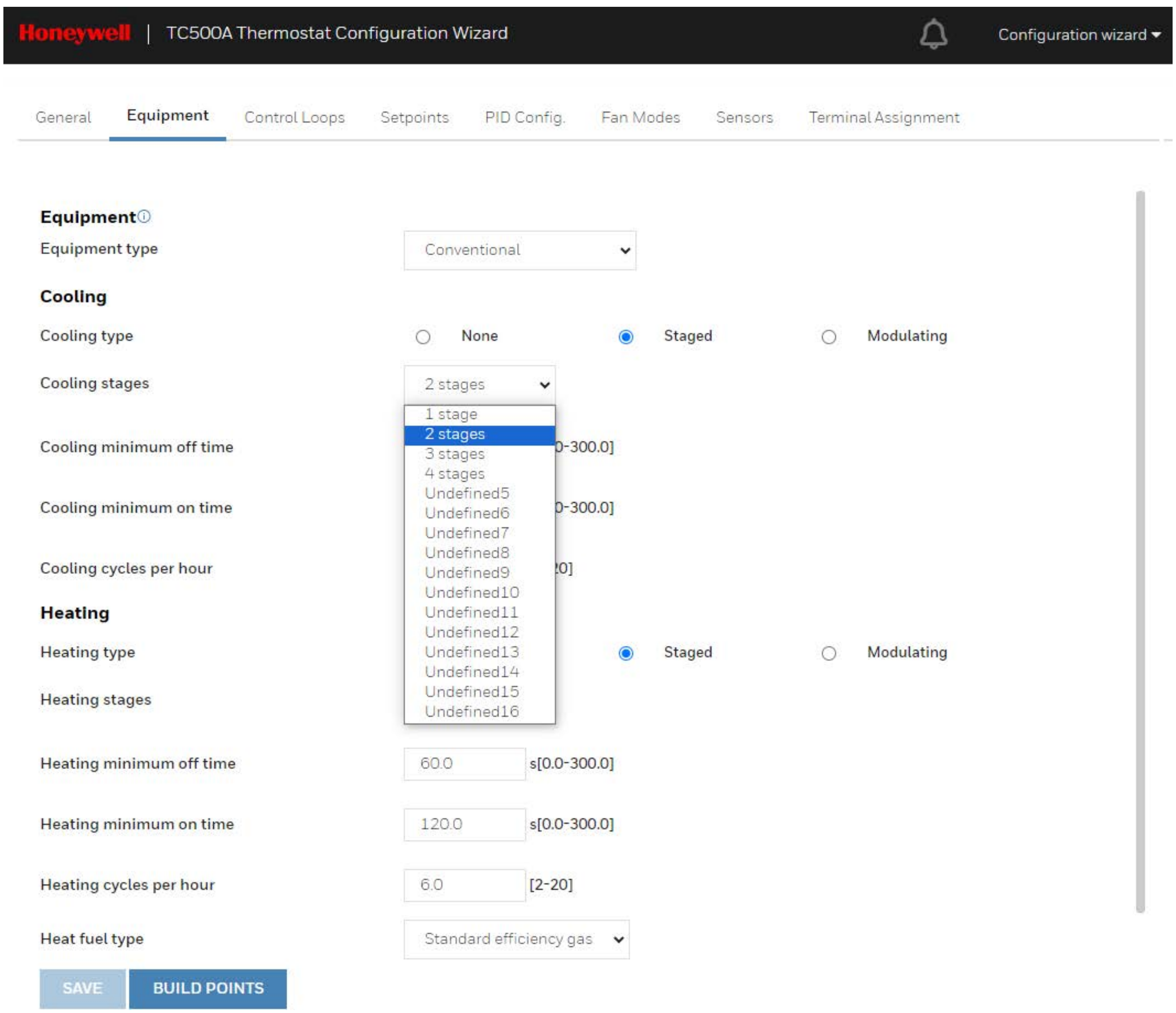

Fig. 6 Equipment Type

EQUIPMENT - TC300

This Menu allows the user to select the equipment type and subtype. Any change in configuration results in generating an unique application selection number. Click the build points button to generate the points associated with the template number.

Equipment Type

Select the required equipment type through this option. It is a primary setting in the configuration as selecting the various other parameters from the different locations depends on the equipment type. The additional settings for the equipment will be enabled or disabled according to the equipment type selected.

The following equipment types are available:

- 4 Pipe Dual Coil
- 4 Pipe Single Coil
- 2 Pipe Single Coil

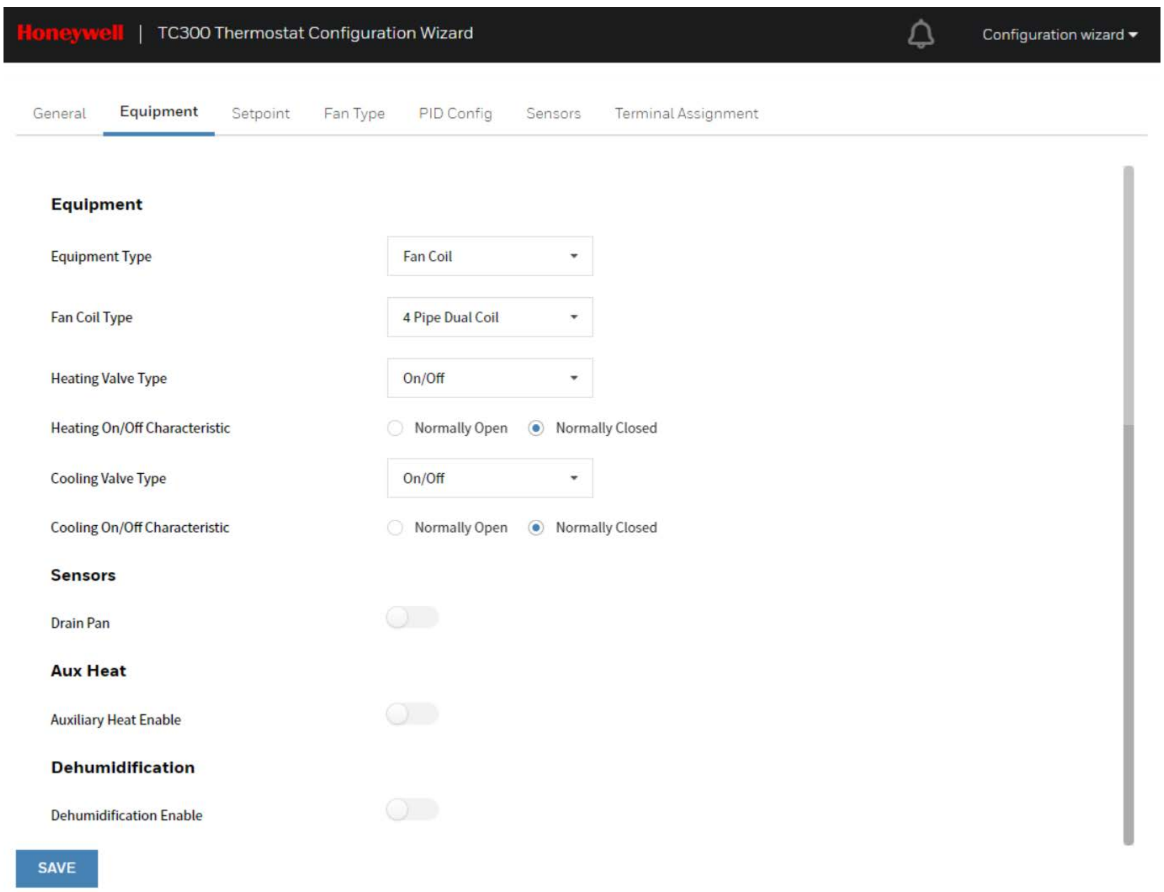

Fig. 7 Equipment Type

CONTROL LOOPS - TC500A

This Menu allows the user to select the Economizer status, Damper positions, CO2 control, Humidification and Dehumidification. Any change in configuration results in heating and cooling. Click the build points button to generate the points associated with the template number.

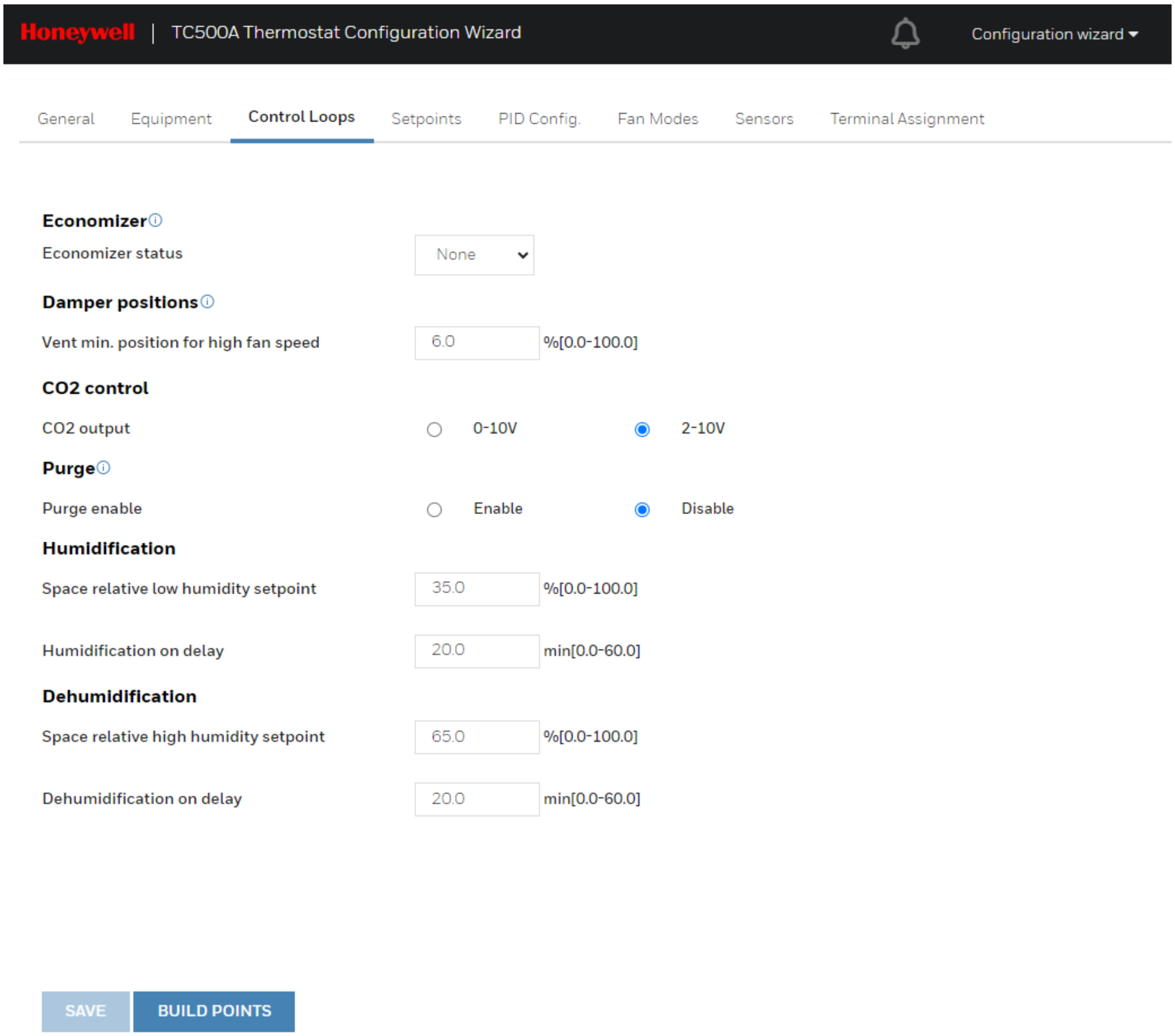

Fig. 8 Control Loops

SETPOINTS - TC500A

This Menu allows the user to configure the setpoints for different modes like occupied, standby, and unoccupied.

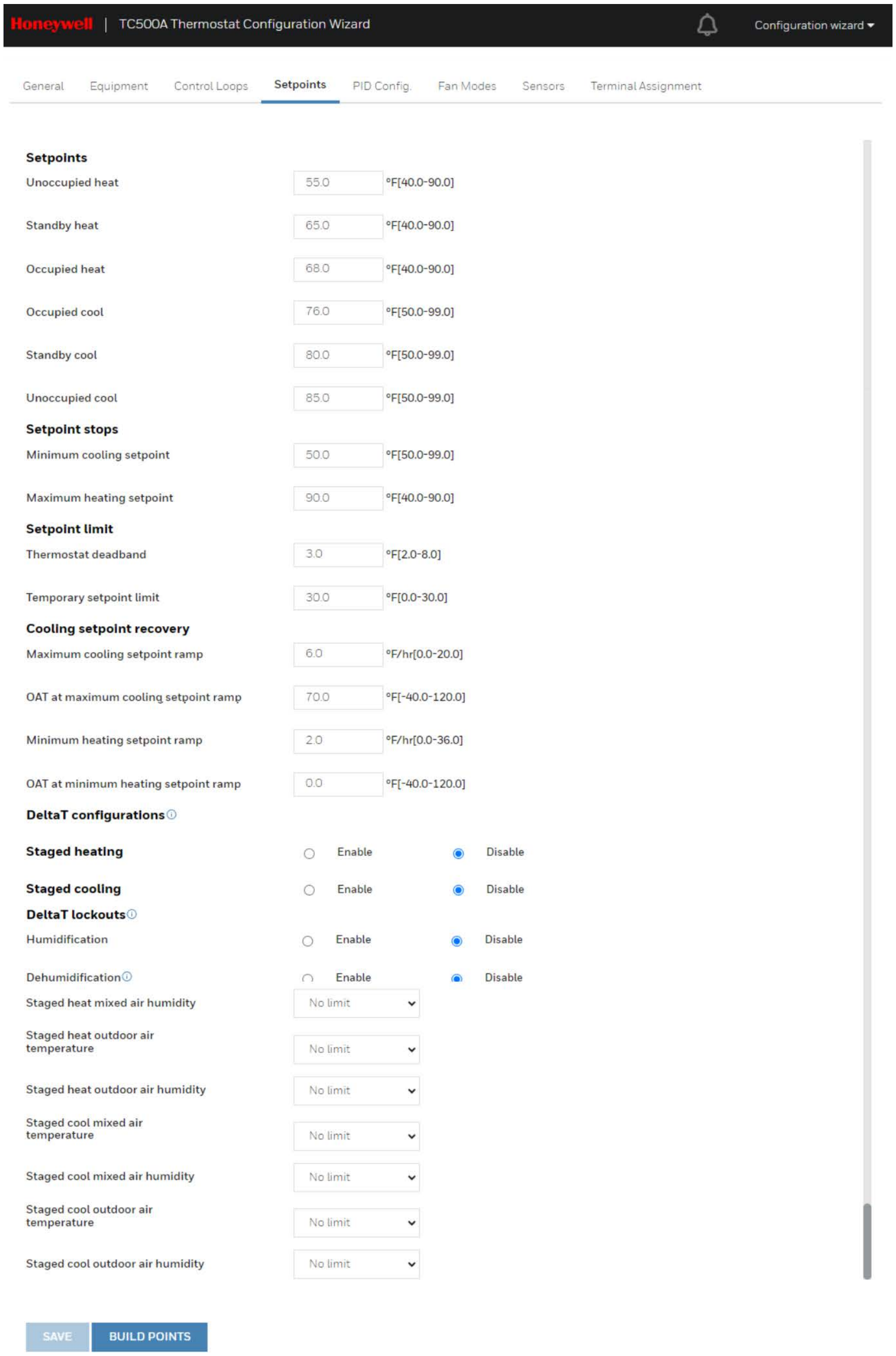

Fig. 9 Setpoints

The space temperature setpoints entered must be as follows:
unoccupied heat setpoint ≤ standby heat setpoint ≤ occupied heat setpoint < occupied cool setpoint ≤ standby cool setpoint ≤ unoccupied cool setpoint.

NOTE:
If a setpoint was entered out of range, when attempting to save the configuration, an error message displays.

SETPOINTS – TC300

This Menu allows the user to configure the setpoints for different modes like occupied, standby, and unoccupied.

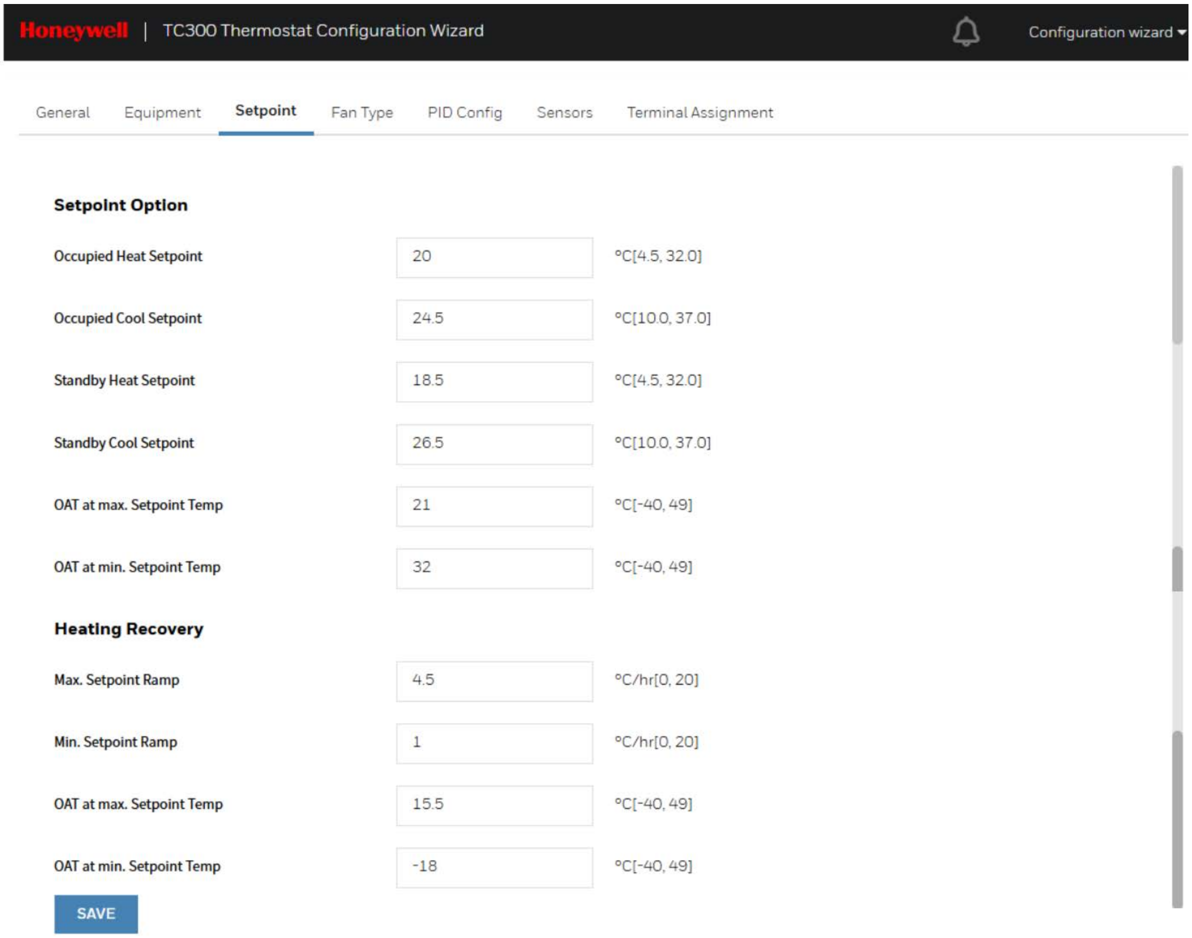

Fig. 10 Setpoints

The space temperature setpoints entered must be as follows:
unoccupied heat setpoint ≤ Occupied Cool setpoint ≤ Standby heat setpoint < Standby cool setpoint ≤ Unoccupied Heat setpoint ≤ unoccupied cool setpoint.

NOTE:
If a setpoint was entered out of range, when attempting to save the configuration, an error message displays.

PID CONFIGURATIONS – TC500A

This Menu allows the user to configure the PID settings for cooling and heating options.

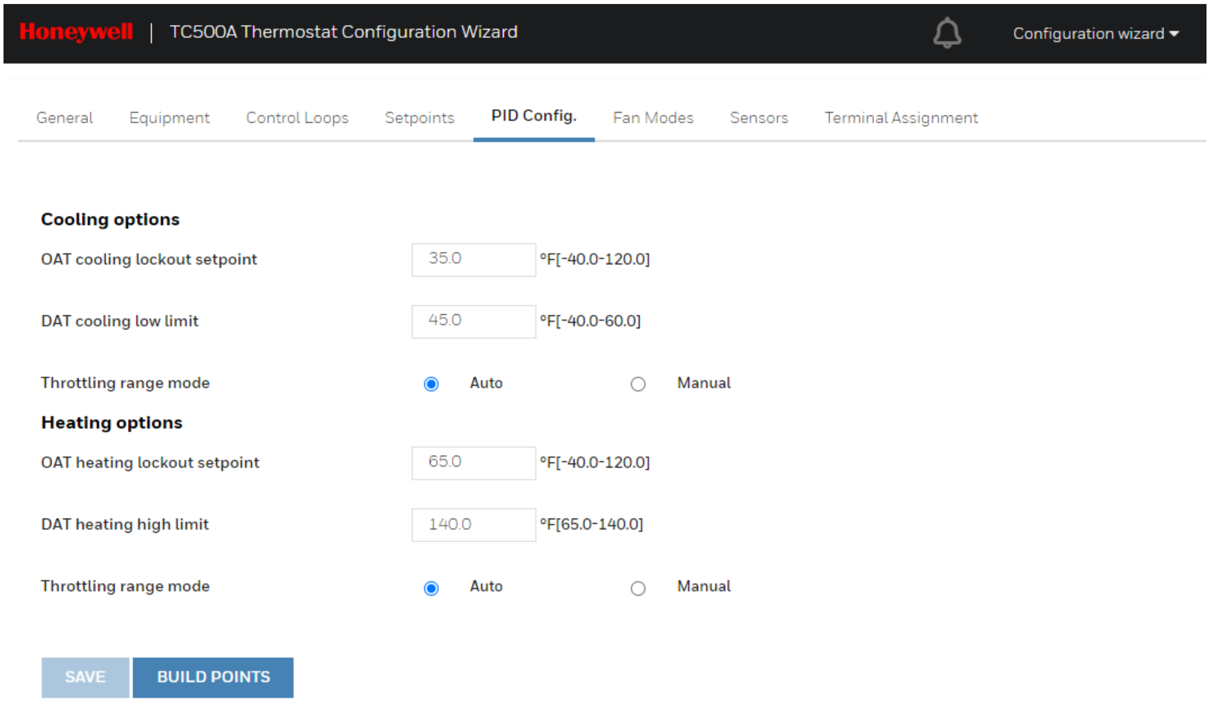

Fig. 11 PID Configurations

PID CONFIGURATIONS – TC300

This Menu allows the user to configure the PID settings for cooling and heating options.

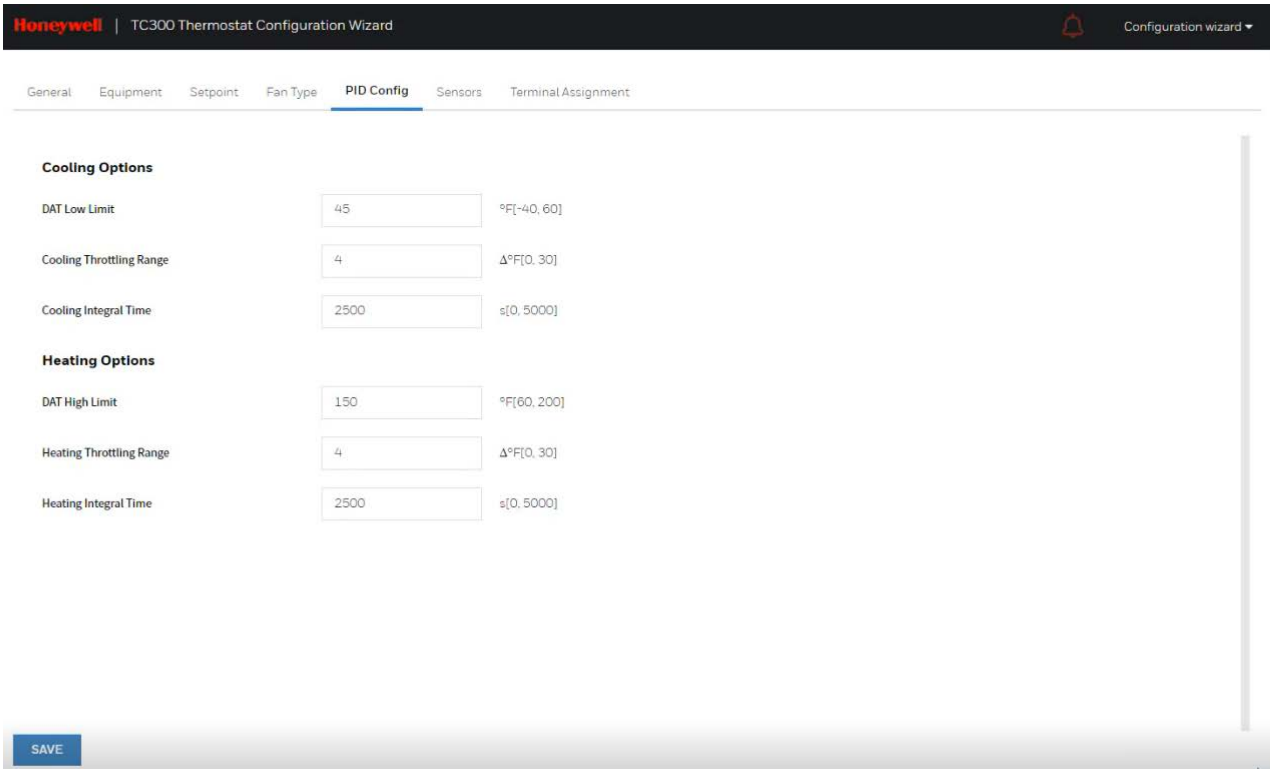

Fig. 12 PID Configurations

FAN MODES - TC500A

The fan modes Menu allows the user to set parameters for the fan. The below settings can be changed.

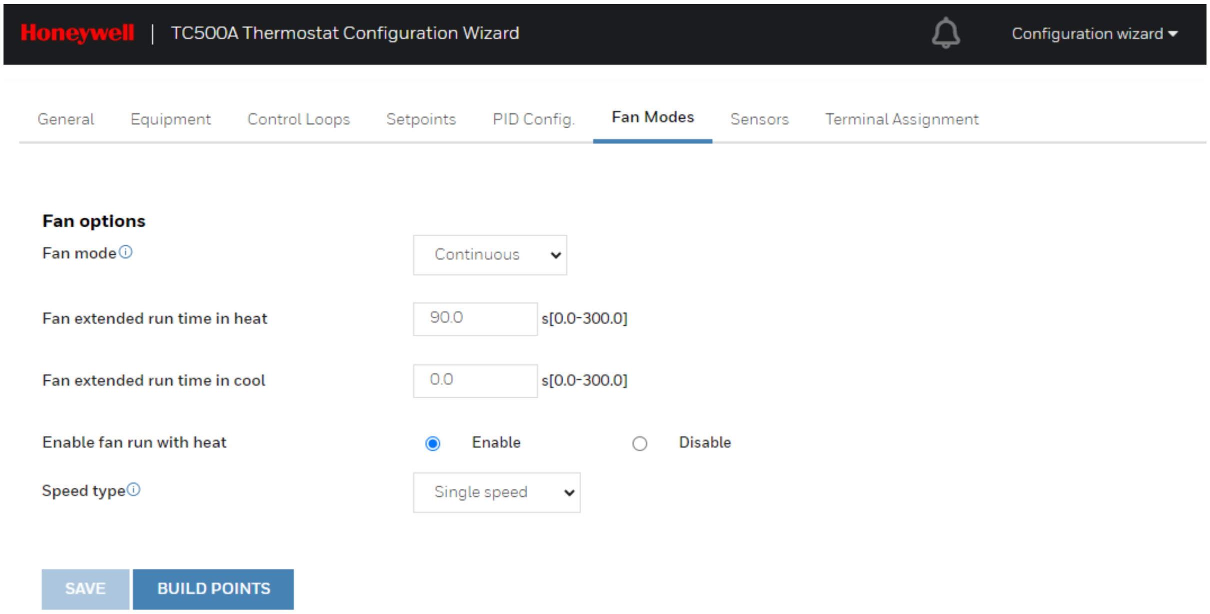

Fig. 13 Fan Modes

FAN TYPE - TC300

The fan modes Menu allows the user to set parameters for the fan. The below settings can be changed.

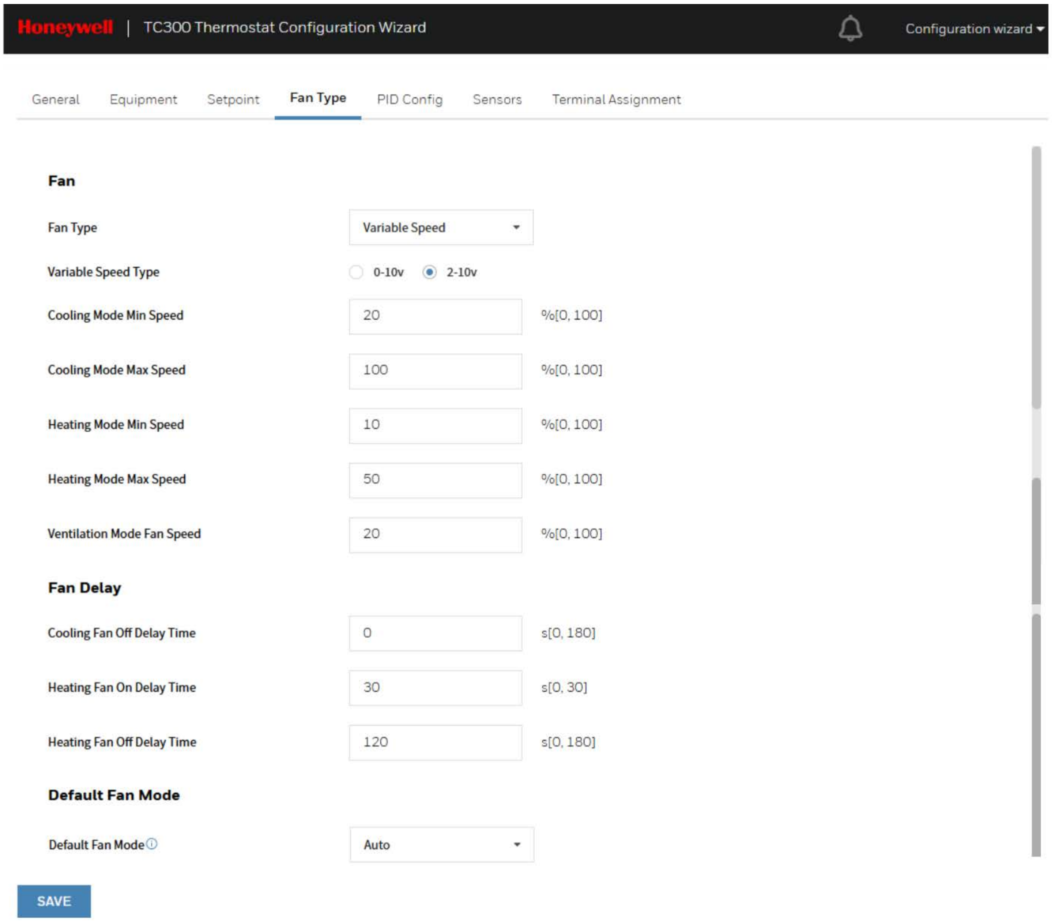

Fig. 14 Fan Modes

NOTE:

Applicable only for TC300 Device.

SENSORS – TC500A

This Menu allows the user to configure the sensors.

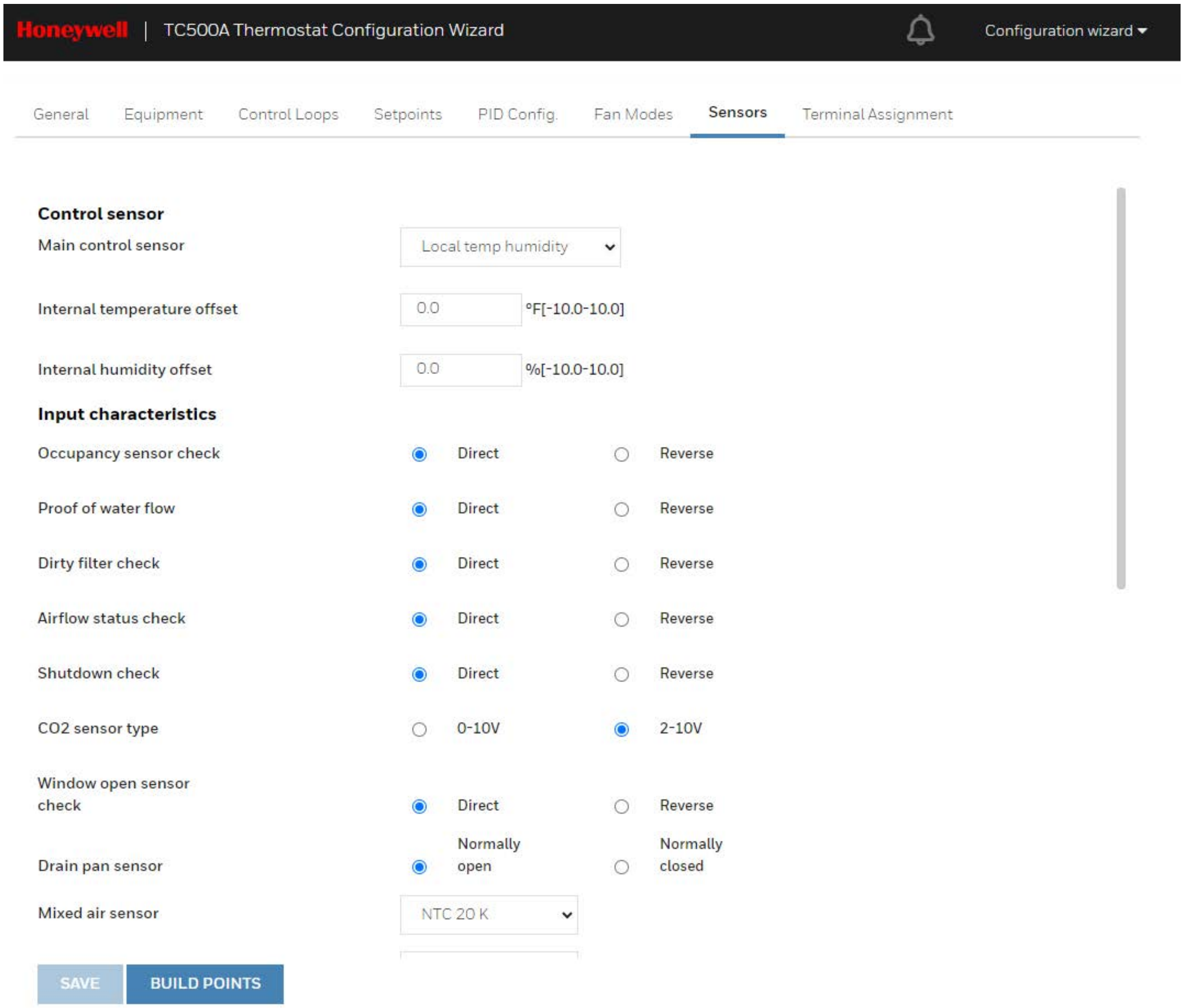

Fig. 15 Sensors

SENSORS – TC300

This Menu allows the user to configure the sensors.

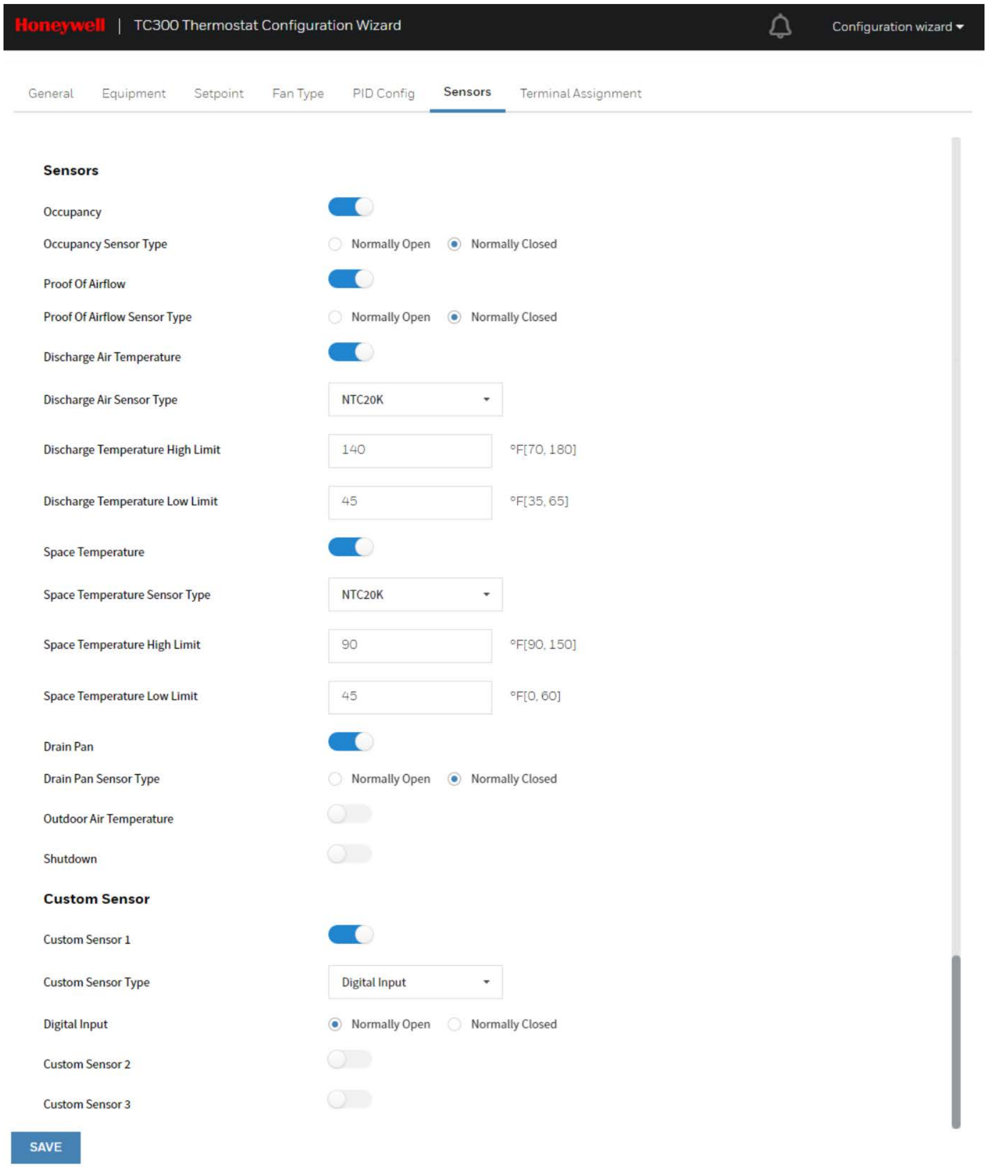

Fig. 16 Sensors

TERMINAL ASSIGNMENT – TC500A

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings.

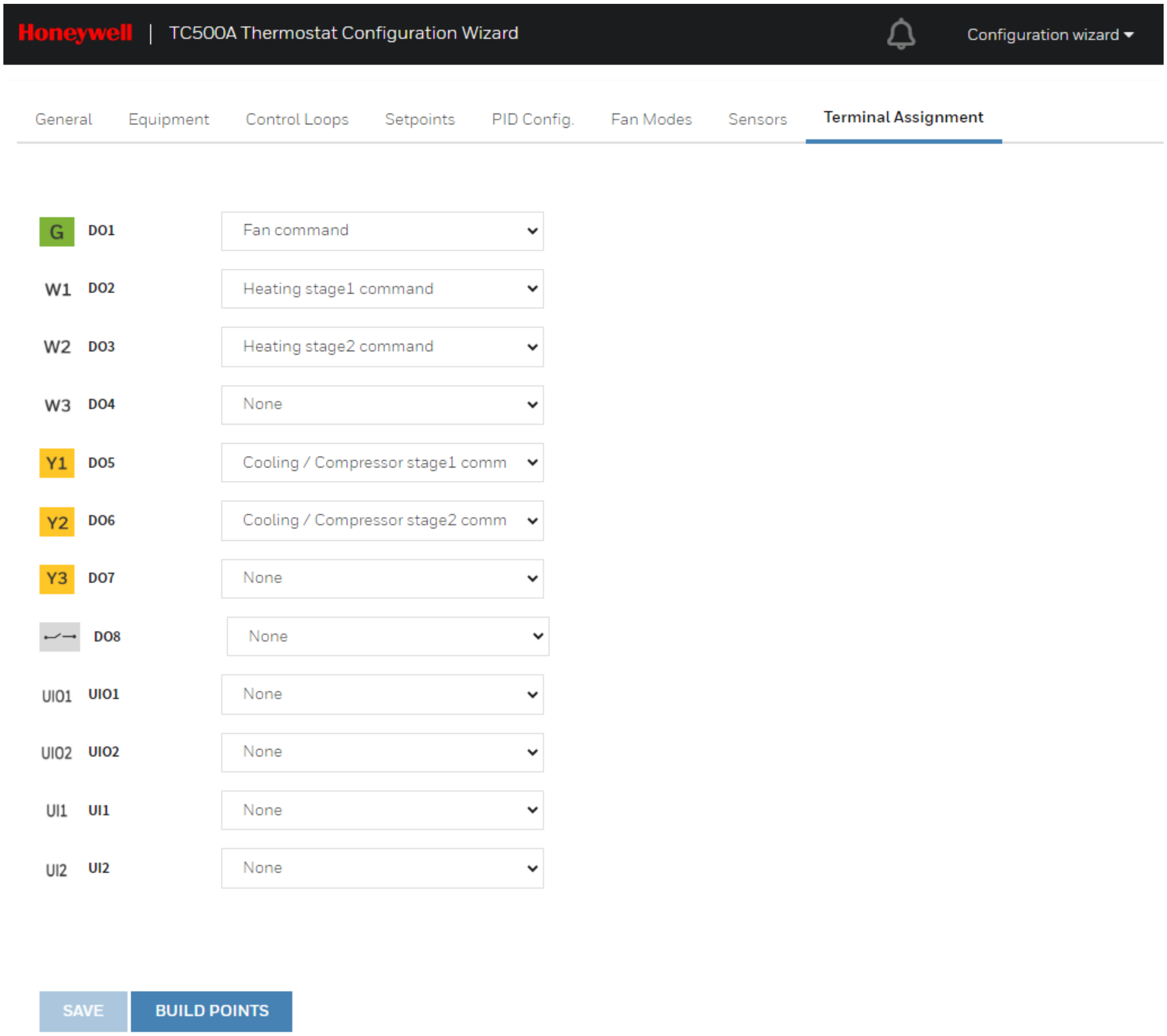

Fig. 17 Terminal Assignment

Table 4 Terminal Assignment

Terminal	Options
DO1	**1:** None **2:** Fan command **(default)** **3:** Fan high-speed command
DO2	**1:** Heating stage1 command **(default)**
DO3	**1:** Heating stage2 command **(default)**
DO4	**1:** None **(default)** **2:** Heating stage3 command **3:** Heat pump reversing valve command **4:** Fan low-speed command **5:** Occupancy status **6:** Dehumidification command **7:** Humidification command **8:** Purge output
DO5	**1:** Cooling / Compressor stage1 command **(default)**
DO6	**1:** Cooling / Compressor stage2 command **(default)**
DO7	**1:** None **(default)** **2:** Cooling / Compressor stage3 command **3:** Economizer min. damper command **4:** Fan low-speed command **5:** Occupancy status **6:** Dehumidification command **7:** Humidification command
DO8	**1:** None **(default)** **2:** Economizer min. damper command **3:** Fan low-speed command **4:** Occupancy status **5:** Dehumidification command **6:** Humidification command **7:** Purge output
UIO1	**1:** None **(default)** **2:** Occupancy sensor **3:** Dirty filter **4:** Air flow status **5:** Shutdown **6:** Mixed air sensor **7:** Outside air sensor **8:** Discharge air sensor **9:** CO2 sensor **10:** Fan speed control **11:** Water flow status **12:** Space temperature sensor **13:** Packaged economizer fault **14:** Window open **15:** Return air sensor **16:** CO2 output **17:** Purge output

Table 4 Terminal Assignment (Continued)

Terminal	Options
UIO2	**1:** None **(default)** **2:** Occupancy sensor **3:** Dirty filter **4:** Air flow status **5:** Shutdown **6:** Mixed air sensor **7:** Outside air sensor **8:** Discharge air sensor **9:** CO2 sensor **10:** Heating control **11:** Water flow status **12:** Space temperature sensor **13:** Packaged economizer fault **14:** Window open **15:** Return air sensor **16:** CO2 output **17:** Purge output
UI1	**1:** None **(default)** **2:** Occupancy sensor **3:** Dirty filter **4:** Air flow status **5:** Shutdown **6:** Mixed air sensor **7:** Outside air sensor **8:** Discharge air sensor **9:** CO2 sensor **10:** Water flow status **11:** Space temperature sensor **12:** Packaged economizer fault **13:** Window open **14:** Return air sensor
UI2	**1:** None **(default)** **2:** Occupancy sensor **3:** Dirty filter **4:** Air flow status **5:** Shutdown **6:** Mixed air sensor **7:** Outside air sensor **8:** Discharge air sensor **9:** CO2 sensor **10:** Water flow status **11:** Space temperature sensor **12:** Heat pump frequency **13:** Packaged economizer fault **14:** Window open **15:** Return air sensor

TERMINAL ASSIGNMENT - TC300

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings.

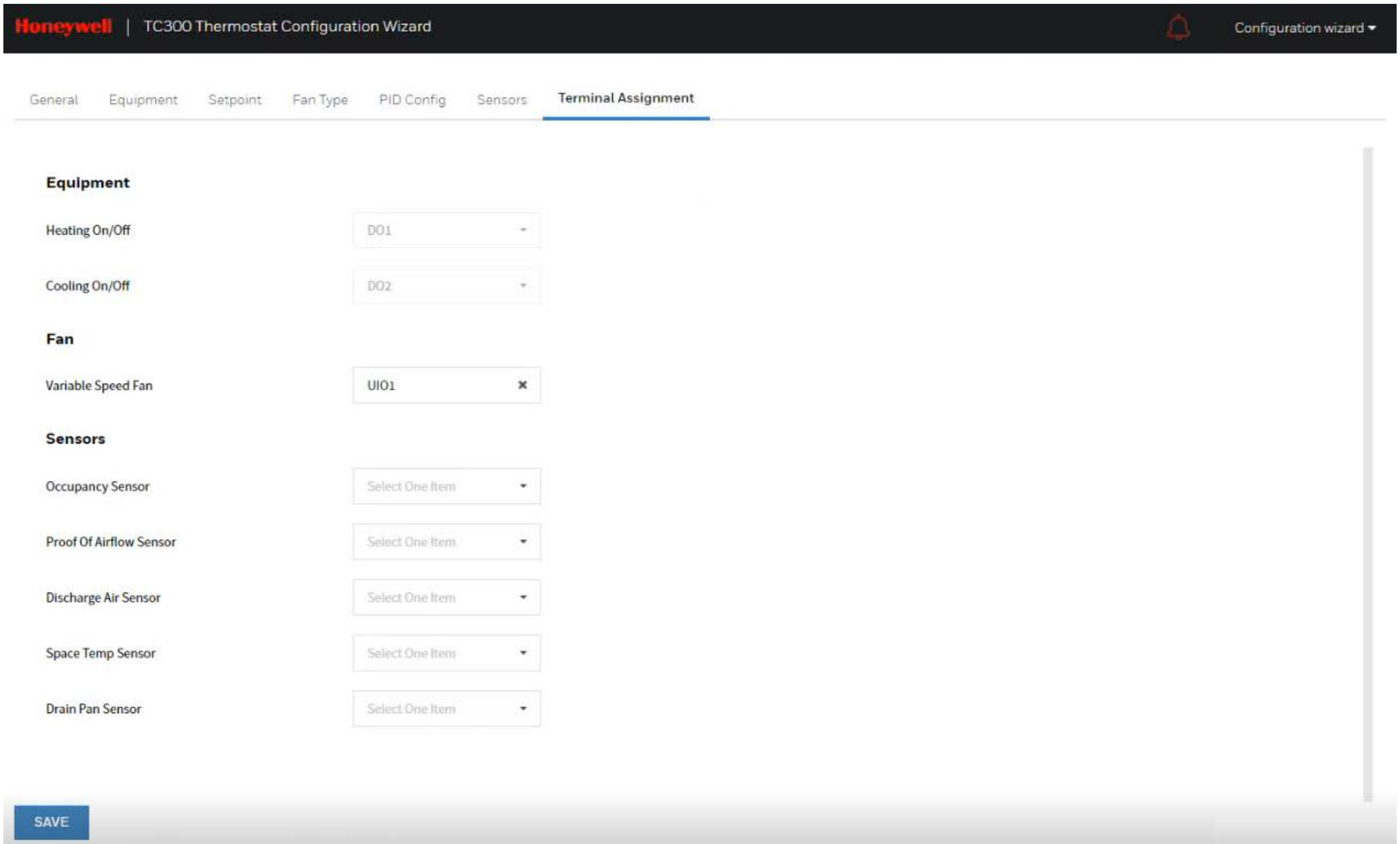

Fig. 18 Terminal Assignment

Table 5 Terminal Assignment

Terminal	Options
DO1	**1:** NA **2:** On/Off Heat **(default)** **3:** Heating On/Off **4:** Heating Floating Open **5:** Cooling Floating Open **6:** Valve On/Off **7:** Valve Floating Open **8:** Changeover Valve **9:**fan Command **10:** High Speed Fan **11:** Medium Speed Fan **12:** Low Speed Fan **13:** Auxiliary Heat **14:** Heat Stage 1 **15:** Valve Stage 1 **Note:** Changeover valve used to switch between heating and cooling mode.
DO2	**1:** NA **2:** On/Off Cool **(default)** **3:** Heating Floating Close **4:** Cool Floating Close **5:** Cooling On/Off **6:** Valve Floating Close **7:** Changeover Valve **8:** Fan Command **9:** High Speed Fan **10:** Medium Speed Fan **11:** Low Speed Fan **12:** Auxiliary Heat **13:** Cool Stage 1
DO3	**1:** NA **2:** NA **(default)** **3:** Cooling Floating Open **4:** Changeover Valve **5:** Fan Command **6:** High Speed Fan **7:** Medium Speed Fan **8:** Low Speed Fan **9:** Auxiliary Heat **10:** Heat Stage 1 **11:** Cool Stage 1

Table 5 Terminal Assignment (Continued)

Terminal	Options
DIO1	**1:** NA **(default)** **Input** **1:** Discharge Air Sensor **2:** Drain Pan Sensor **3:** Occupancy Sensor **4:** Proof of Airflow **5:** Pipe Sensor **6:** Space Temp Sensor **7:** Changeover Switch **Output** **1:** Cooling Floating Close **2:** Changeover Valve **3:** Fan Command **4:** High Speed Fan **5:** Medium Speed Fan **6:** Low Speed Fan **7:** Auxiliary Heat
DIO2	**1:** NA **(default)** **Input** **1:** Discharge Air Sensor **2:** Drain Pan Sensor **3:** Occupancy Sensor **4:** Proof of Airflow **5:** Pipe Sensor **6:** Space Temp Sensor **7:** Changeover Switch **Output** **1:** Changeover Valve **2:** Fan Command **3:** High Speed Fan **4:** Medium Speed Fan **5:** Low Speed Fan **6:** Auxiliary Heat
UIO1	**1:** NA **(default)** **Input** **1:** Discharge Air Sensor **2:** Drain Pan Sensor **3:** Occupancy Sensor **4:** Proof of Airflow **5:** Pipe Sensor **6:** Space Temp Sensor **7:** Changeover Switch **Output** **1:** 6-Way Valve **2:** Modulating Cool **3:** Modulating Heat **4:** Modulating Valve **5:** UIO1 UIO2 NA Variable Speed Fan

Table 5 Terminal Assignment (Continued)

Terminal	Options
UIO2	**1:** NA **(default)** **Input** **1:** Discharge Air Sensor **2:** Drain Pan Sensor **3:** Occupancy Sensor **4:** Proof of Airflow **5:** Pipe Sensor **6:** Space Temp Sensor **7:** Changeover Switch **Output** **1:** 6-Way Valve **2:** Modulating Cool **3:** Modulating Heat **4:** Modulating Valve **5:** UIO1 UIO2 NA Variable Speed Fan
UIO3	**1:** NA **(default)** **Input** **1:** Discharge Air Sensor **2:** Drain Pan Sensor **3:** Occupancy Sensor **4:** Proof of Airflow **5:** Pipe Sensor **6:** Space Temp Sensor **7:** Changeover Switch **Output** **1:** 6-Way Valve **2:** Modulating Cool **3:** Modulating Heat **4:** Modulating Valve **5:** UIO1 UIO2 NA Variable Speed Fan

UPLOAD/DOWNLOAD CONFIGURATION

The TC Series Thermostat Configuration Wizard allows the user to download and upload the configuration to the TC Series device after configuration changes.

Download Configuration

This action downloads the saved configuration from the wizard to the TC500A device. After every configuration change, it is essential to download the configuration to the device to reflect the new settings. To download the updated configuration to the TC Series thermostat device, right-click on the TC500 device, click **Actions**, and click **Download Configuration**

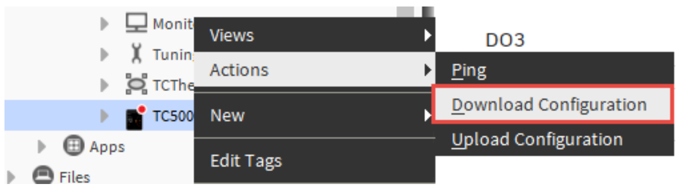

Fig. 19 Download Configuration

Download Configuration (UX Screen)

Double click on Configuration Download

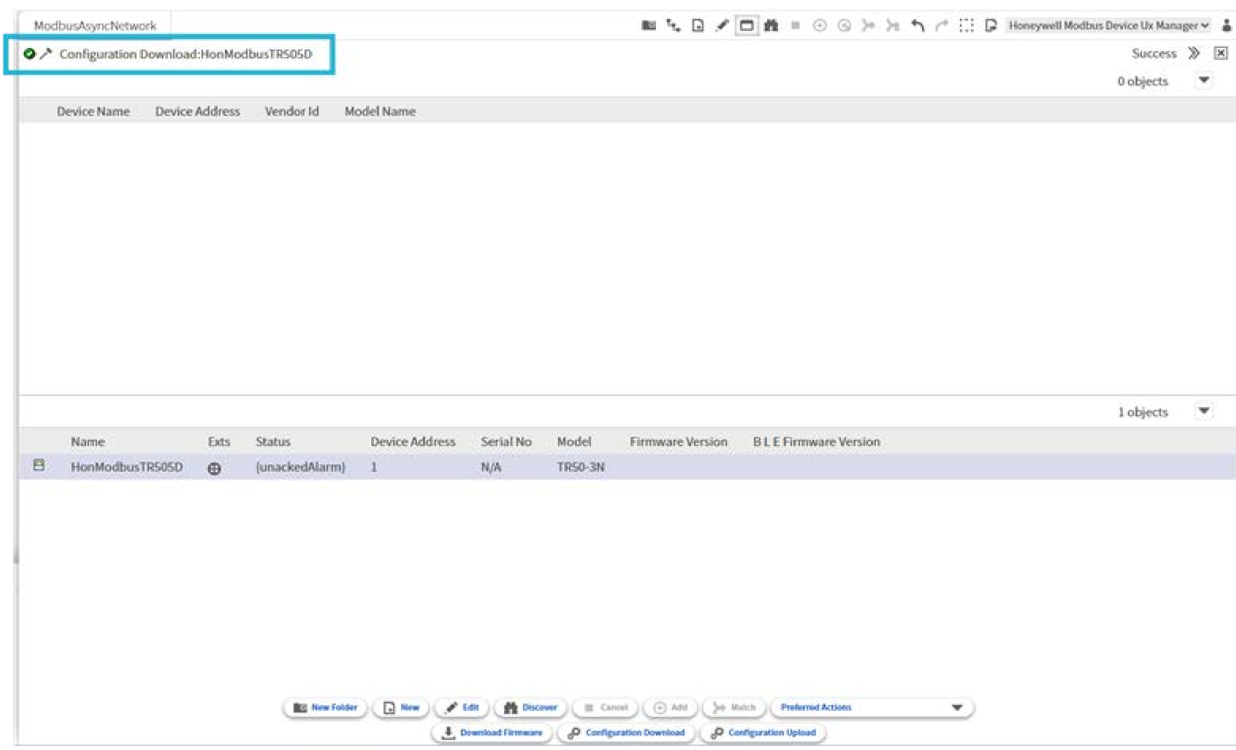

Fig. 20 Download Configuration (UX screen)

Upload Configuration

This action allows the user to upload the configuration saved in the TC Series device. After uploading, settings can be reconfigured and downloaded to the device again. To upload the existing configuration from the TC Series device, click **Actions** and then click **Upload Configuration**.

Fig. 21 Upload Configuration

Upload Configuration (UX Screen)

Double click on Configuration Download .

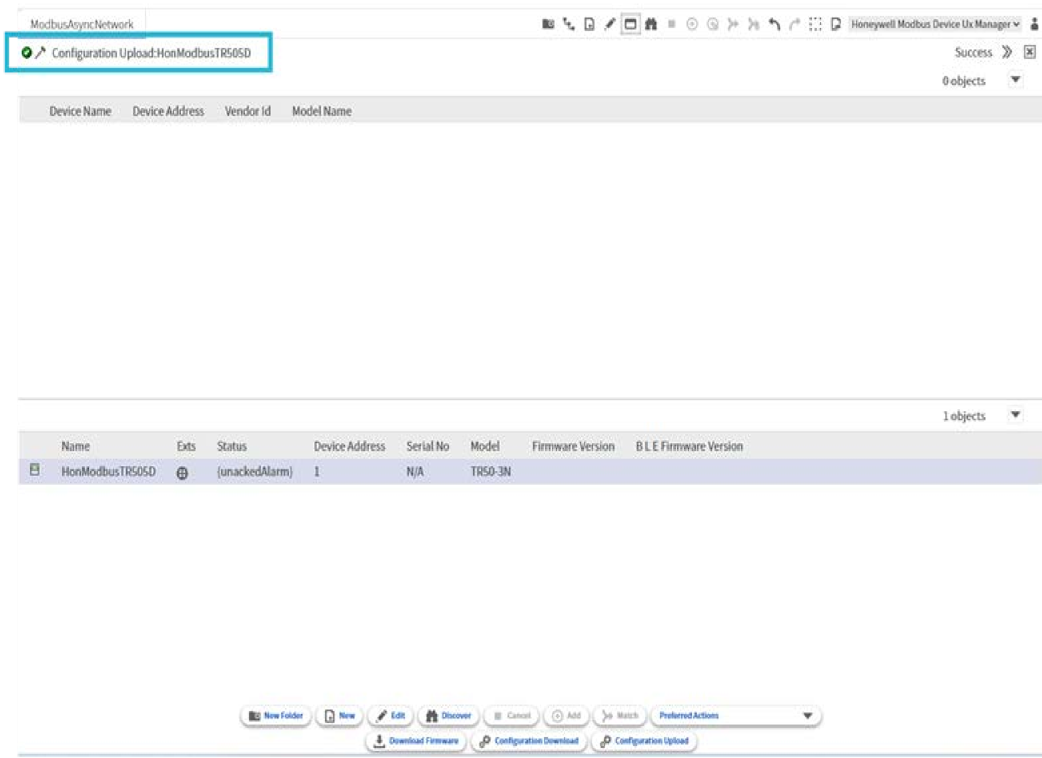

Fig. 22 Upload Configuration (UX Screen)

NOTE:
The upload and download configuration will support undefined values only when the thermostat has valid enum ordinal.

SERVICE MODE – TC500A

Service mode allows the user to test the output of the equipment for the mechanical operation.

Click on the drop-down menu and select the service mode to navigate to the service mode page. The service mode includes the analog and digital outputs of the TC500A thermostat. The output points display the outputs available on the thermostat, and the override option allows to configure a value and set the output on the thermostat.

To use the service mode:

1. Configure the toggle switch to ON to enable the override value fields.

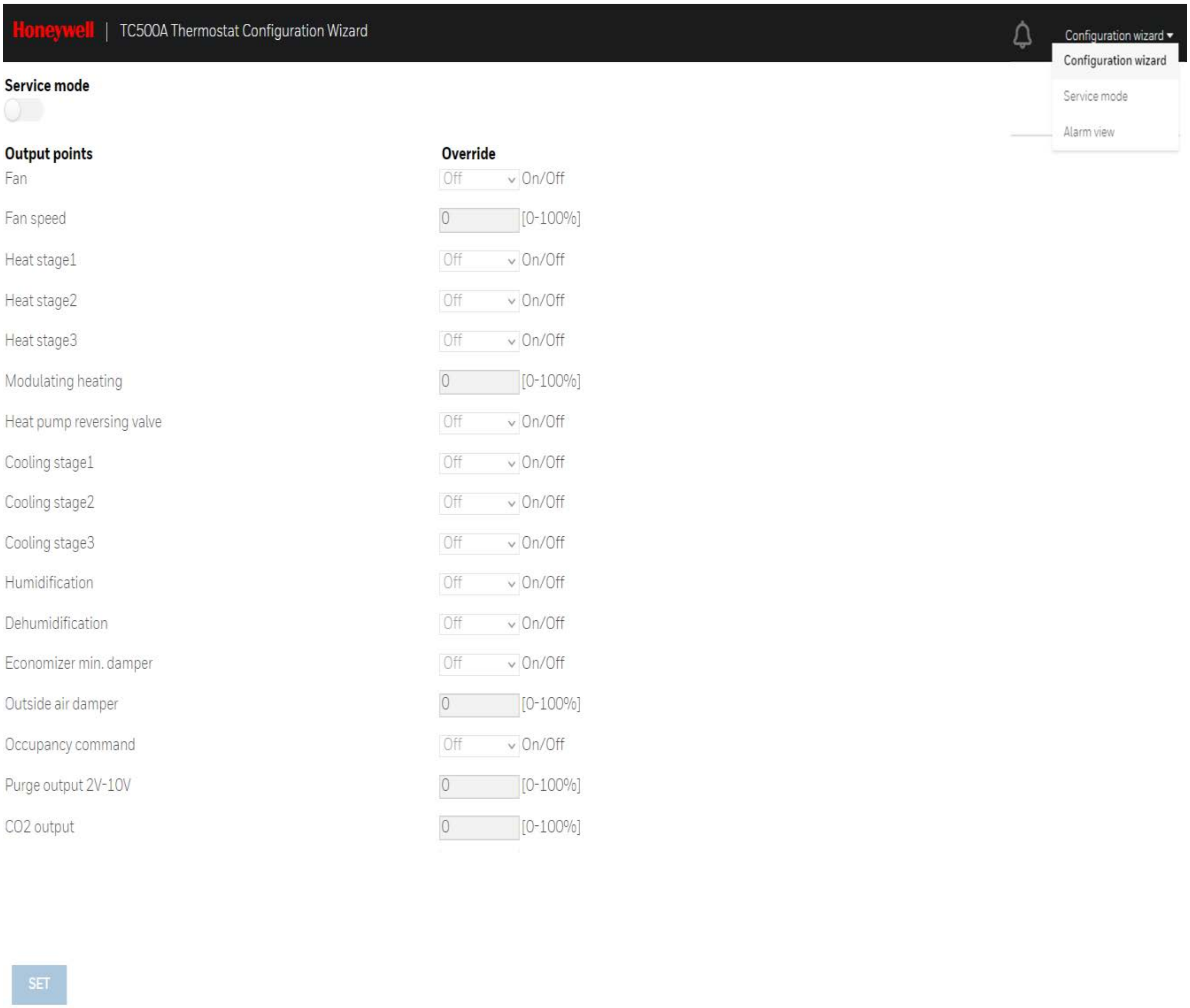

Fig. 23 Service Mode – Disabled

2. Click **Yes** on the notification displayed to enable the service mode.

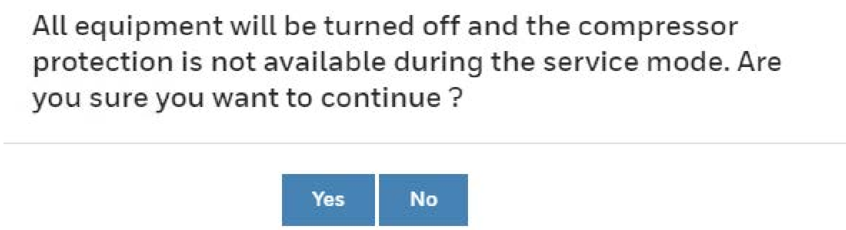

Fig. 24 Dialogue Box to Enable the Service Mode

3. Configure the required values in the override fields and click on the set button to set the values to the output on the thermostat.

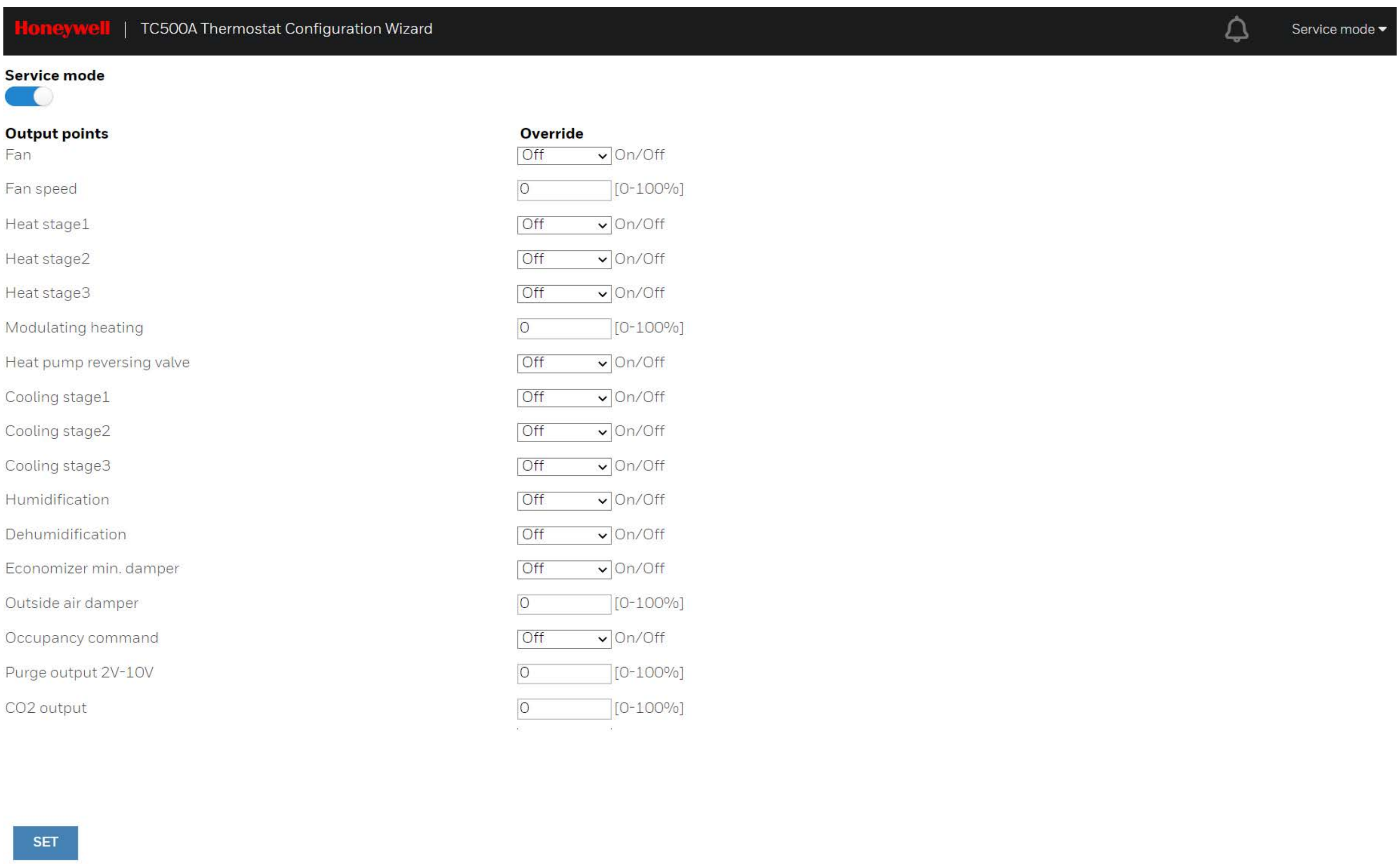

Fig. 25 Service Mode – Enabled

1. Disable the toggle switch to disable the service mode.
2. Click on the drop-down menu and select the configuration wizard to navigate to the configuration wizard view. A prompt to exit the service mode will appear. Select Yes, if you want to exit the service mode and navigate to the configuration wizard. Select No, if you want to continue with the overridden points and navigate to the configuration wizard. Select cancel to continue with the service mode and not navigate to the configuration wizard.

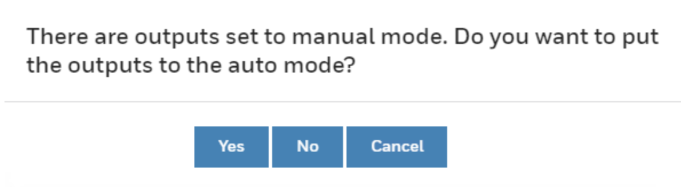

Fig. 26 Dialogue Box to Disable the Service Mode

NOTE:

The service mode is displayed only when the device is online. The outputs are written on the network inputs. Use the respective network output to validate the point value.

The override is for a one-hour duration and restarts whenever a point is overridden using set option. After one-hour, the thermostat will exit service mode.

SERVICE MODE – TC300

Service mode allows the user to test the output of the equipment for the mechanical operation.

Click on the drop-down menu and select the service mode to navigate to the service mode page. The service mode includes the analog and digital outputs of the TC300 thermostat. The output points display the outputs available on the thermostat, and the override option allows to configure a value and set the output on the thermostat.

To use the service mode:

1. Configure the toggle switch to ON to enable the override value fields.

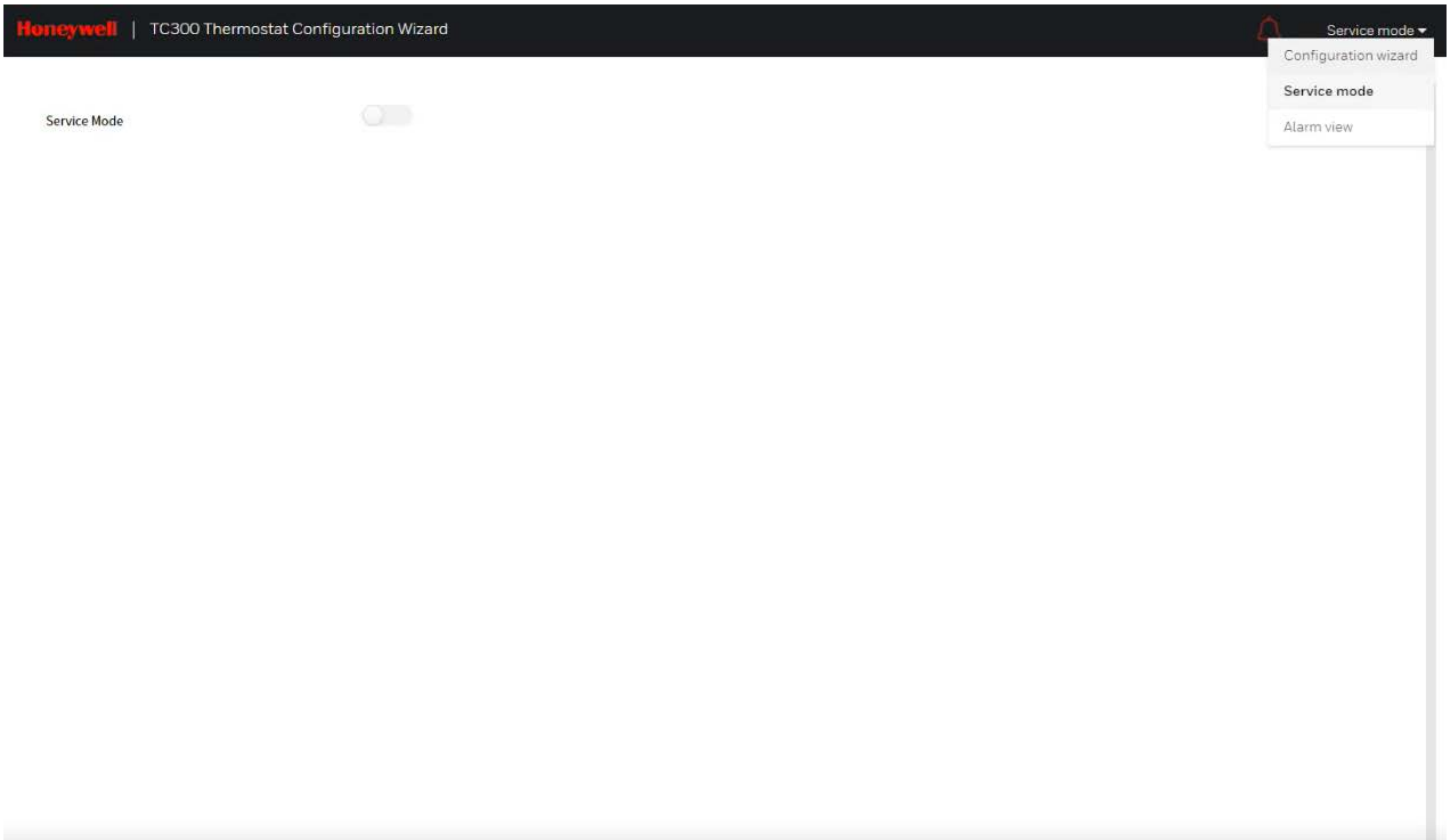

Fig. 27 Service Mode – Disabled

2. Click **Yes** on the notification displayed to enable the service mode.

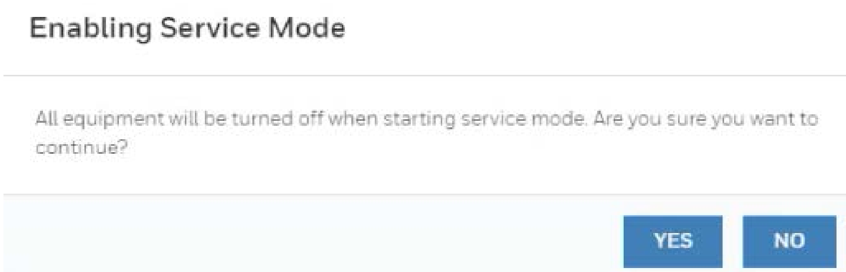

Fig. 28 Dialogue Box to Enable the Service Mode

3. Configure the required values in the override fields and click on the set button to set the values to the output on the thermostat.

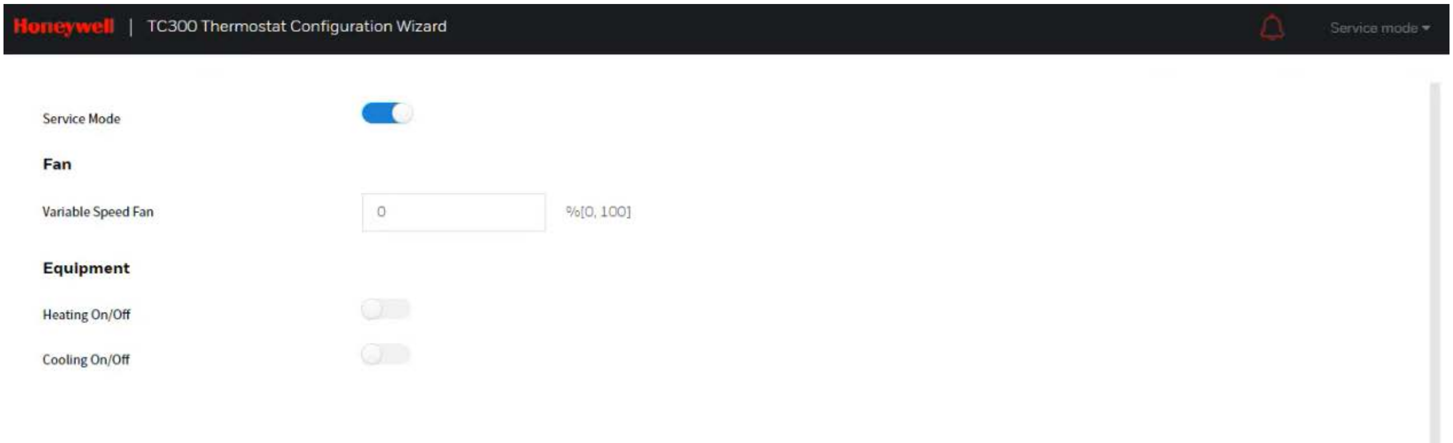

Fig. 29 Service Mode - Enabled

1. Disable the toggle switch to disable the service mode.
2. Click on the drop-down menu and select the configuration wizard to navigate to the configuration wizard view. A prompt to exit the service mode will appear. Select Yes, if you want to exit the service mode and navigate to the configuration wizard. Select No, if you want to continue with the overridden points and navigate to the configuration wizard. Select cancel to continue with the service mode and not navigate to the configuration wizard.

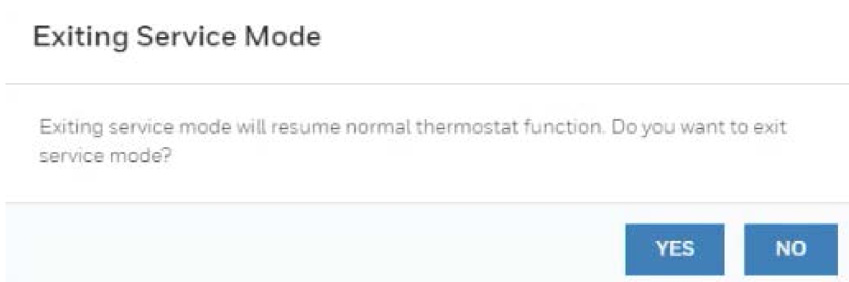

Fig. 30 Dialogue Box to Disable the Service Mode

NOTE:

The service mode is displayed only when the device is online. The outputs are written on the network inputs. Use the respective network output to validate the point value.

The override is for a one-hour duration and restarts whenever a point is overridden using set option. After one-hour, the thermostat will exit service mode.

ALARM VIEW – TC500A

The alarm view allows the user to monitor the predefined alarms in the TC500A thermostat. Navigate to the alarm view page from the navigation tree of the wizard. A red color alarm icon on the header indicates an active alarm in the thermostat, and a white color alarm icon indicates no alarm. The alarm priority is displayed for a point in the alarm. Hover over the tooltip to view the alarm description.

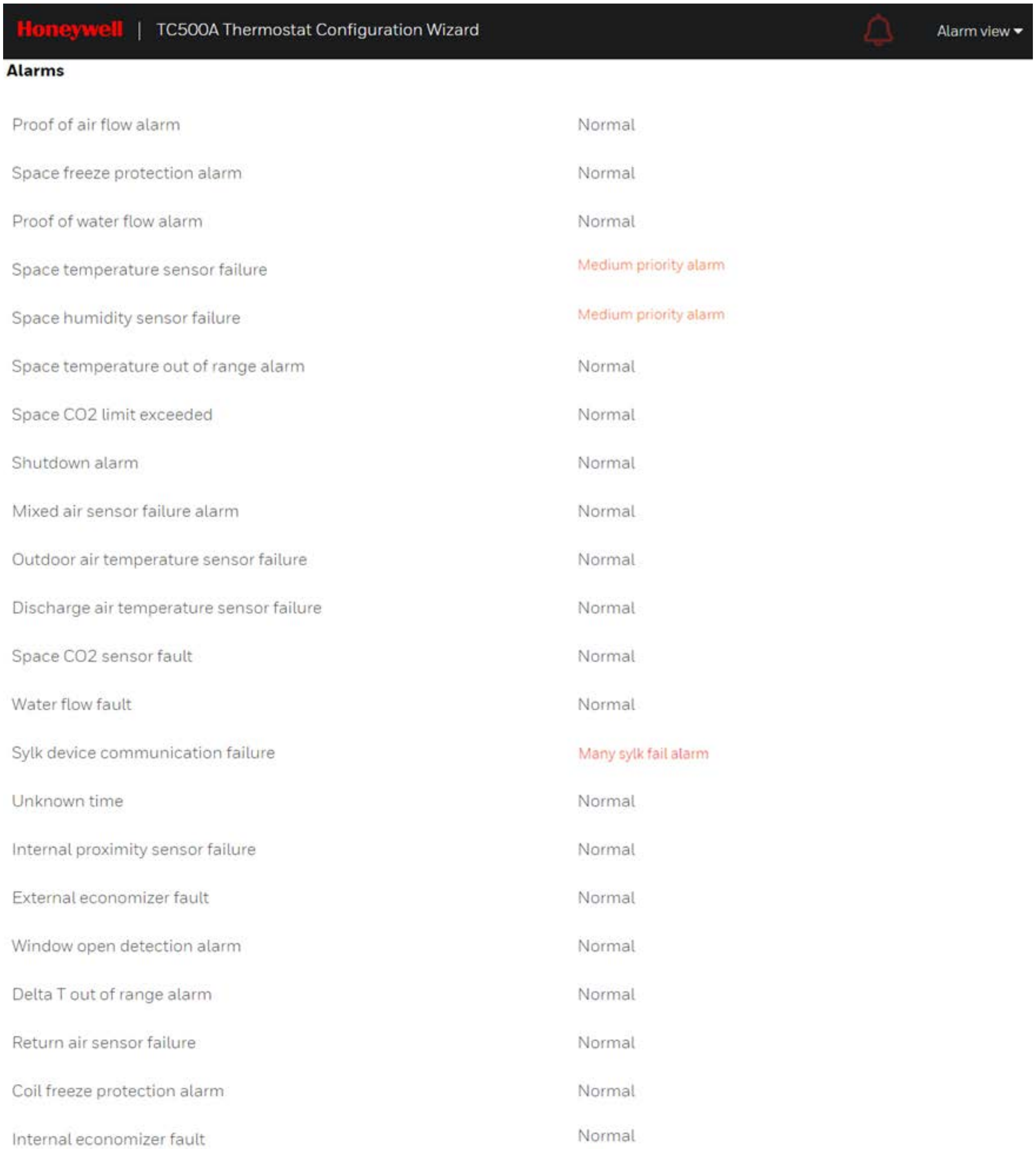

Fig. 31 Alarm View

The following points are configured in the alarm view.

Table 6 Alarm View

Alarm Parameters	Alarm Priority
Proof of air flow alarm	High priority alarm
Space freeze protection alarm	High priority alarm
Proof of water flow alarm	High priority alarm
Space temperature sensor failure	Medium priority alarm High priority alarm
Space humidity sensor failure	Medium priority alarm High priority alarm
Space temperature out of range	Medium priority alarm
Space CO2 limit exceeded	Medium priority alarm (CO2 threshold limit exceeded)
Shutdown alarm	Medium priority alarm
Mixed air sensor failure alarm	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Outdoor air temperature sensor failure	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Discharge air temperature sensor failure alarm	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Space CO2 sensor fault	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Water flow fault	Medium priority alarm
Sylk device communication failure	Sylk address 2 fail alarm Sylk address 3 fail alarm Sylk address 4 fail alarm Sylk address 5 fail alarm Sylk address 6 fail alarm Unconfigured Sylk address 8 fail alarm Sylk address 9 fail alarm Sylk address 10 fail alarm Sylk address 11 fail alarm Sylk address 12 fail alarm Many Sylk fail alarm
Unknown time	High priority alarm
Internal proximity sensor failure	High priority alarm
External economizer fault	High priority alarm
Window open detection alarm	Medium priority alarm
Delta T out of range alarm	Medium priority alarm High priority alarm
Return Air sensor failure	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Coil freeze protection alarm	High priority alarm

Table 6 Alarm View (Continued)

Alarm Parameters	Alarm Priority
Internal economizer fault	Damper stuck open alarm Damper stuck at minimum alarm Bad or unplugged actuator alarm Actuator mechanically disconnected alarm
Drain pan sensor alarm	Normal alarm High Priority alarm

ALARM VIEW - TC300

The alarm view allows the user to monitor the predefined alarms in the TC300 thermostat. Navigate to the alarm view page from the navigation tree of the wizard. A red color alarm icon on the header indicates an active alarm in the thermostat, and a white color alarm icon indicates no alarm. The alarm priority is displayed for a point in the alarm. Hover over the tool tip to view the alarm description.

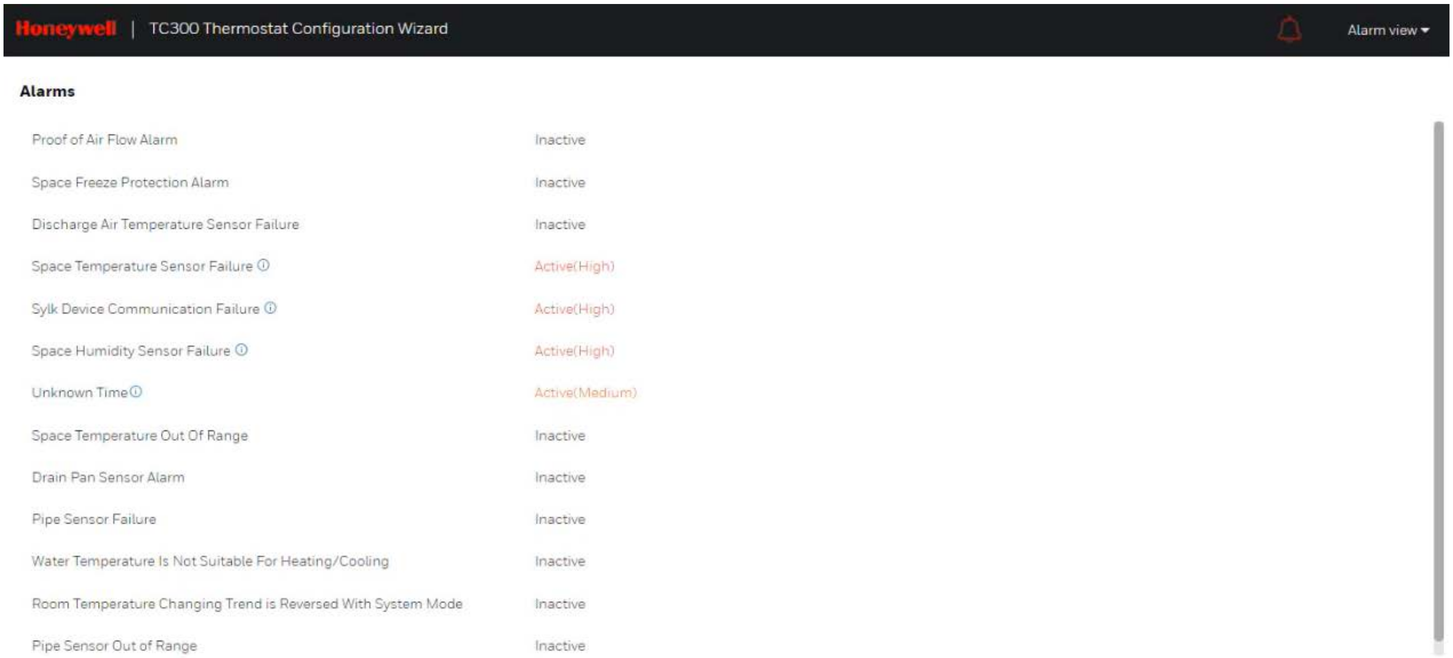

Fig. 32 Alarm View

List of Alarms and their Severity

The list of alarms in the Commercial Connected thermostat is as follows

Table 7 List of Alarm and their Severity

Alarms	Severity
Proof of Air Flow Alarm	High
Space Freeze Protection Alarm	High
Space Temperature Sensor Failure	High/Medium
Space Temperature Out of Range	High/Medium
Space Humidity Sensor Failure	High/Medium
Discharge Air Temp. Sensor Failure	High
Discharge Air Temperature out of range alarm	Medium
Drain Pan Sensor Alarm	High
Sylk Device Communication Failure	High
Pipe Sensor Failure	High
Pipe Sensor Out of Range	High

Table 7 List of Alarm and their Severity (Continued)

Alarms	Severity
Water Temperature is Not Suitable for Heating/Cooling	High
Room Temperature Changing Trend is Reversed with System Mode	High
Unknown Time	Medium

FIRMWARE DOWNLOAD TC SERIES

NOTE:
TC500A and TC300 Configuration are same but the TC500A configuration is shown as reference.

1. Expand **Station** > **Config** > **Driver**.
2. Right-click on the BacnetNetwork > Views > T C Thermostat Bacnet Device Manager.
   The list of all available Bacnet devices which are in the database appears.

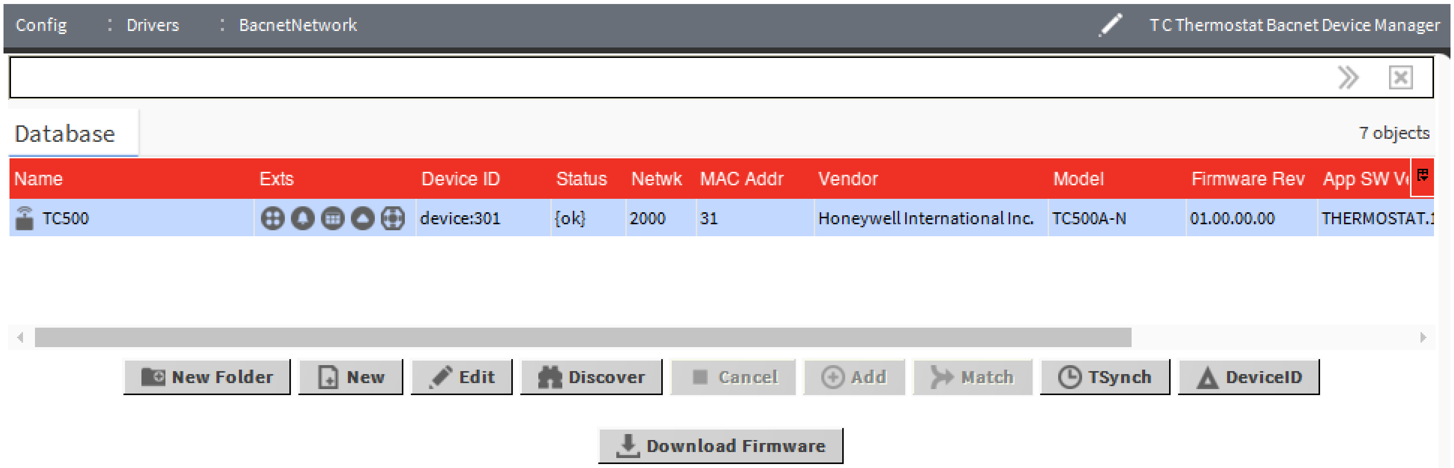

Fig. 33 Firmware Download

3. Select the TC500A thermostat device (Device type = honeywellTCThermostatWizard:TC500).
4. Click on the **Download Firmware** button. The file chooser window appears.
5. Select the firmware file and click on the **Open** button.
   The firmware download starts, and the user will get a notification after the download process is completed.

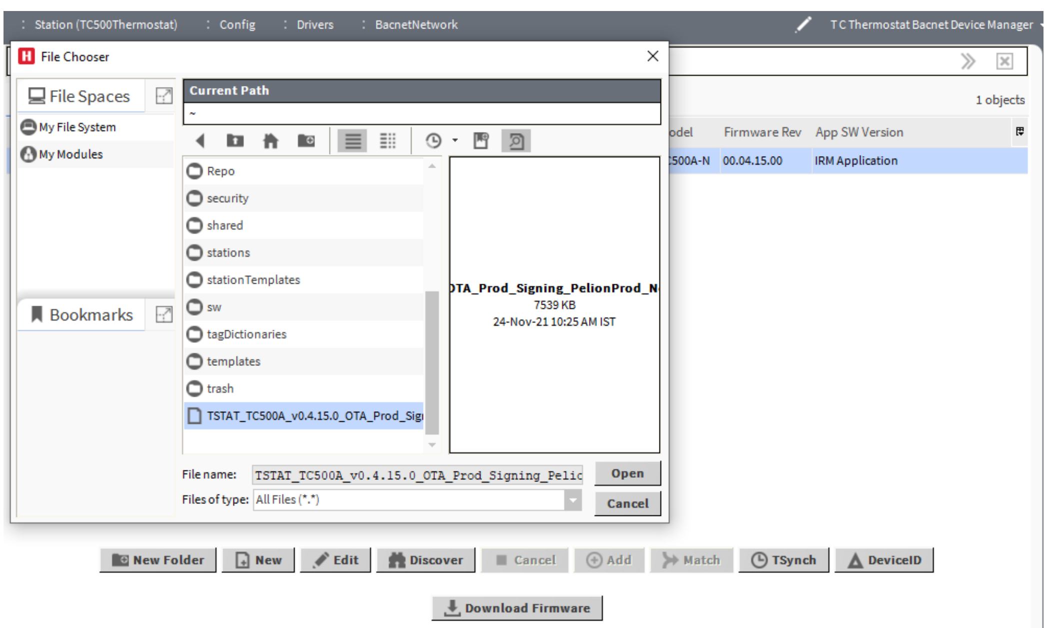

Fig. 34 File Chooser

6. Select the firmware file and click on ok button.

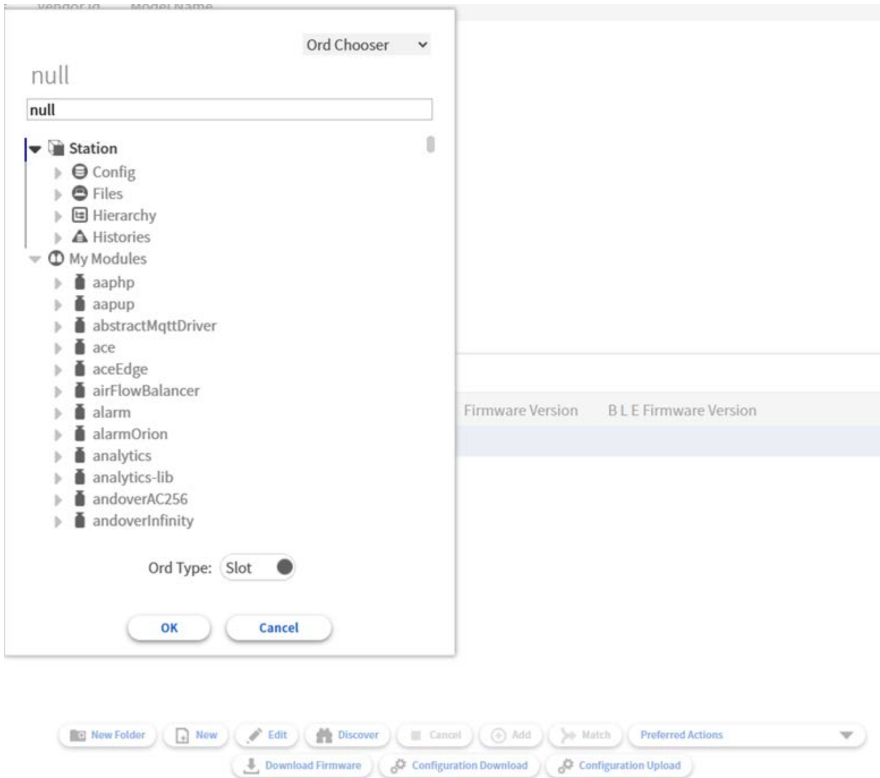

Fig. 35 File Chooser (UX Screen)

NOTE:
The firmware download is supported only through the workbench. Refresh the device manager view page when the firmware download is finished to verify if the firmware version number has been updated.

APPLICABLE TECHNICAL LITERATURE

Table 8 Applicable Technical Literature

Document Name	Document Number
TC500A Thermostat BACnet Integration Guide	31-00478
TC Thermostat Configuration Wizard Installation Instructions	31-00489
TC500A Commercial Thermostat Configuration and User Guide	31-00400M
TC300 Thermostat BACnet Integration Guide	31-00646

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